Results And Disucssion

System Design and Development

Initial EEG Data Collection Using TGAM



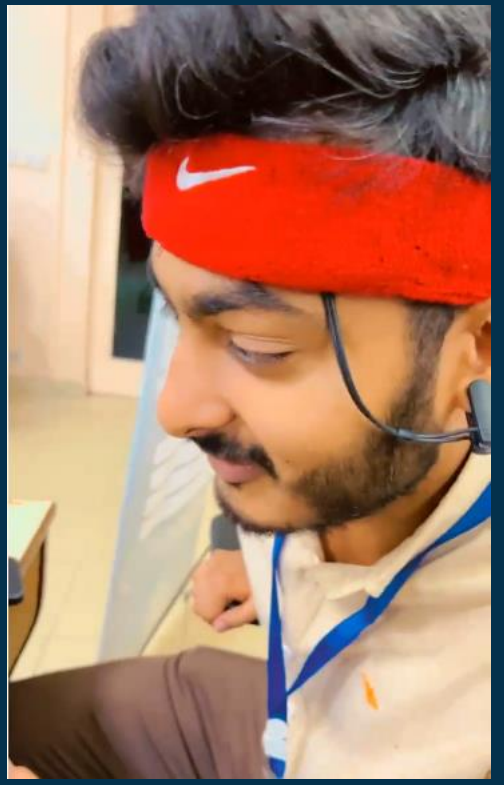


System Design and Development

The EEG system is designed using Band and TGAM for data processing



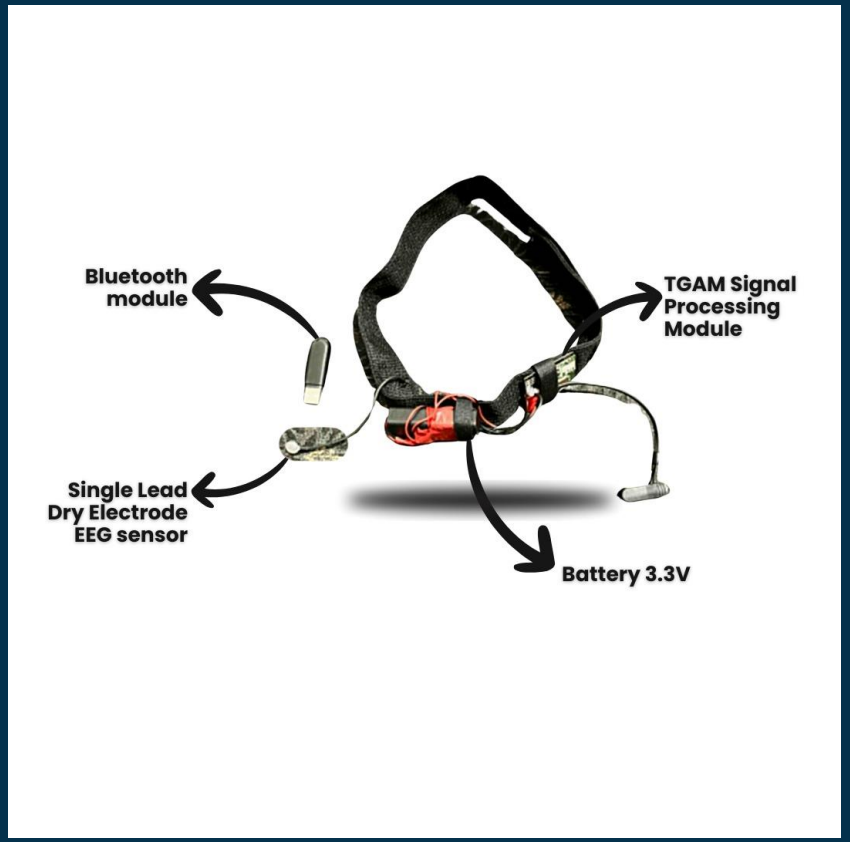




System Design and Development

The system is designed by using EEG technology, TGAM for data processing





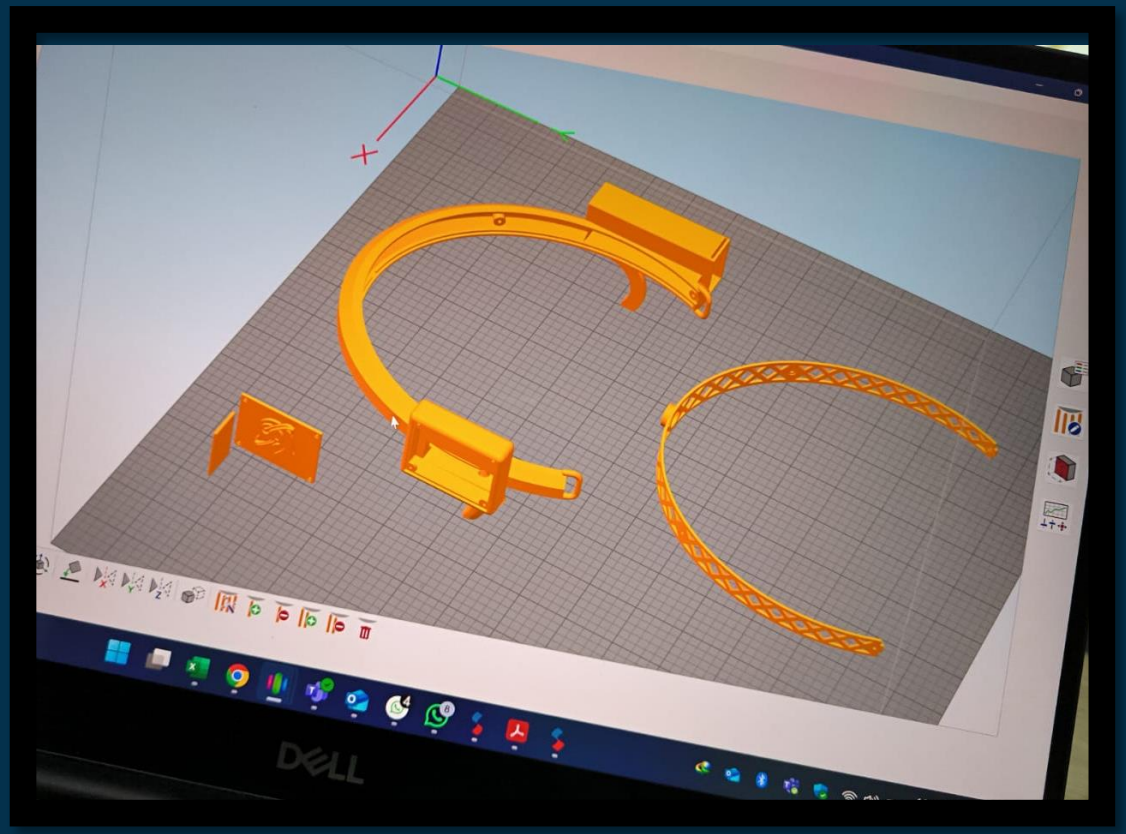
System Design and Development

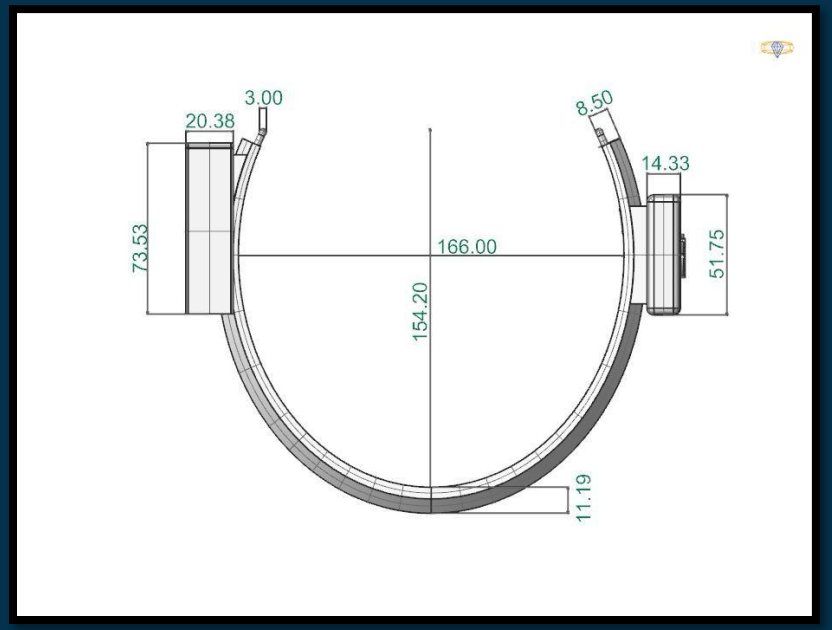
Realtime EEG Data Collection of Student in Class.



System Design and Development

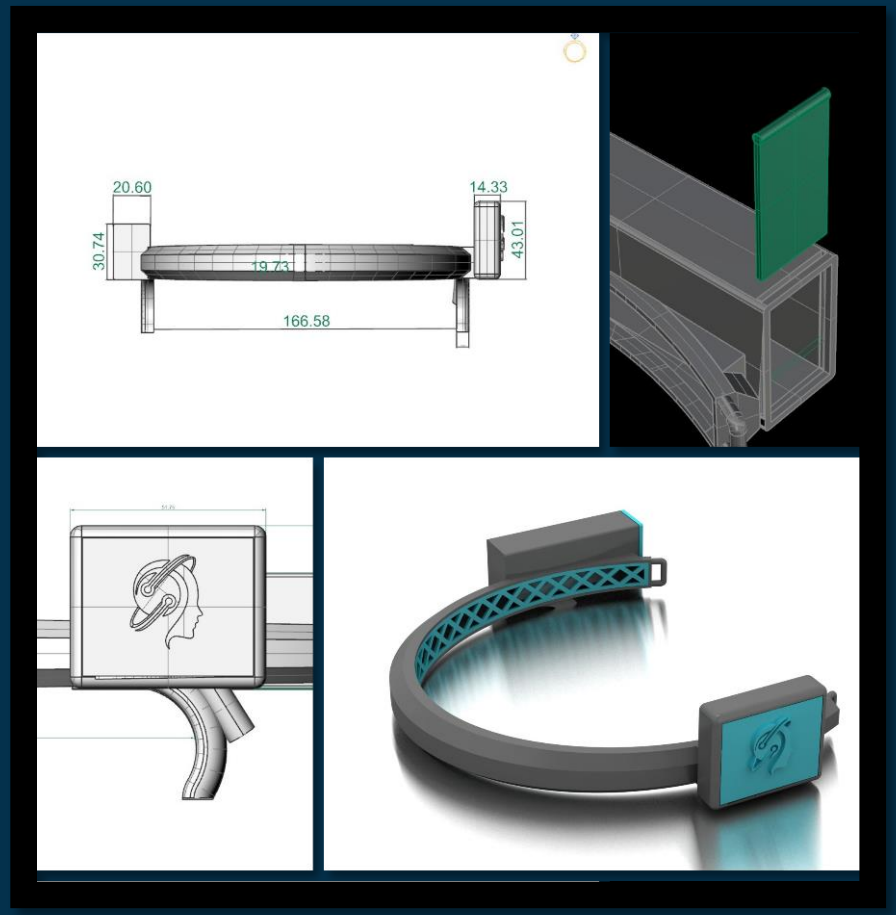
3D Modeling and Architecture of the TGAM Module for EEG Acquisition and Cognitive State Monitoring

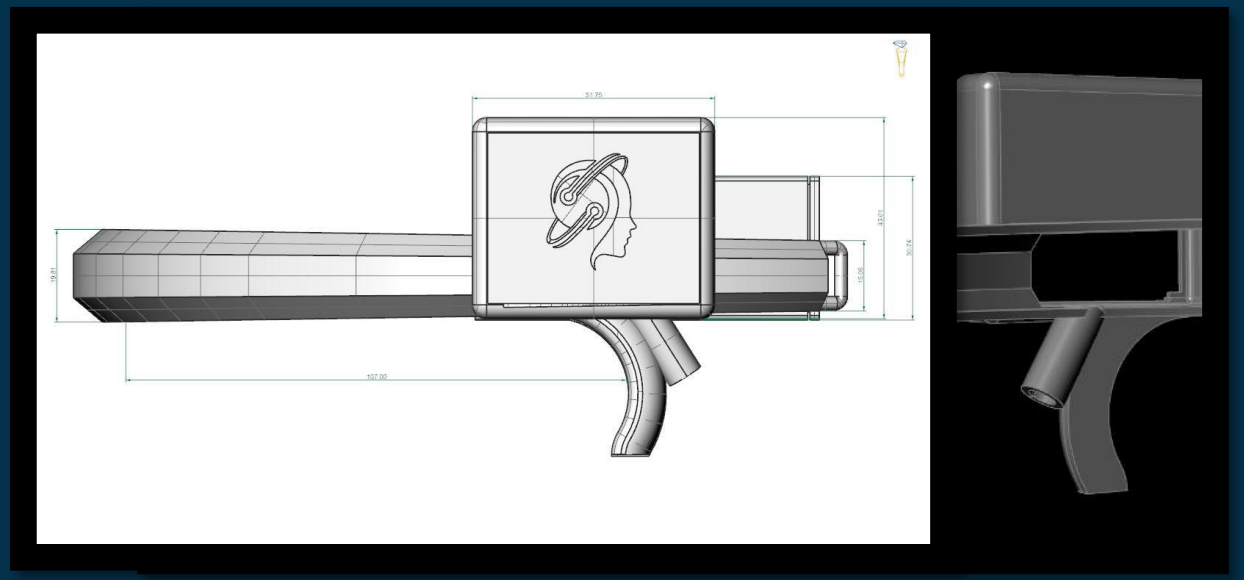




System Design and Development

3D Modeling and Architecture of the TGAM Module for EEG Acquisition and Cognitive State Monitoring

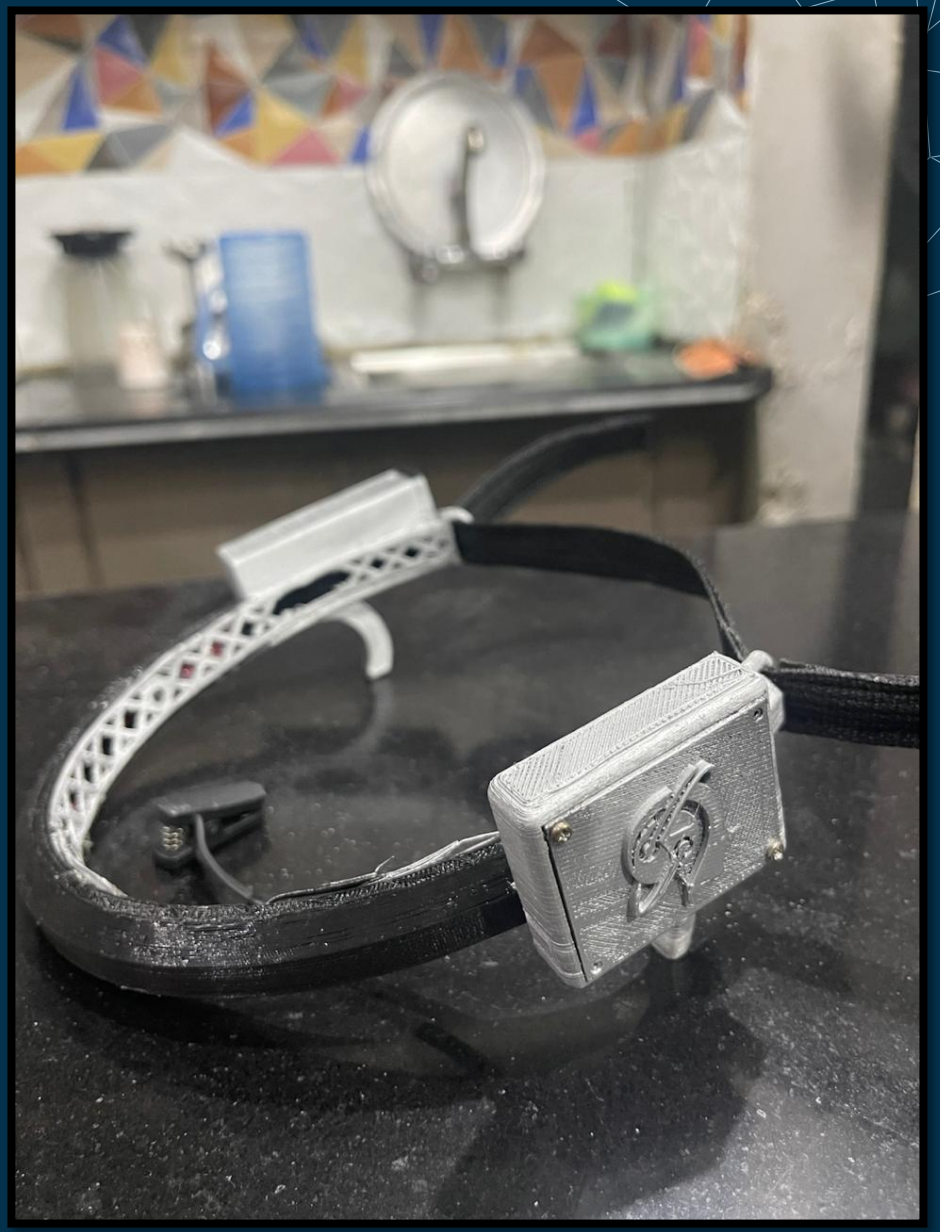




System Design and Development

Hardware Integrated in Headband





System Testing

Students Wearing Headband in Classes





EEG Data From TGAM Module

```python
import serial
import time  # Import the time module

# Establish serial connection to COM4
ser = serial.Serial('COM4', 57600)  # Adjust baud rate as needed

try:
    while True:
        # Read a line of ASCII data from the serial port
        ascii_data = ser.readline().decode('utf-8', errors='ignore').strip()

        # Convert ASCII data to decimal values
        decimal_data = [ord(char) for char in ascii_data]

        # Debugging: Print raw data and decimal values
        print("Raw Data:", ascii_data)
        print("EEG Data For 5Sec:", decimal_data)

        # Introduce a delay to control the sampling rate
        time.sleep(5)  # Adjust the delay (in seconds) as needed

except KeyboardInterrupt:
    print("Keyboard interrupt detected. Exiting gracefully.")

# Close serial connection when done
ser.close()
```

```python
def plot_fft(fft_freq, fft_magnitude):
    plt.plot(fft_freq, fft_magnitude)
    plt.title('EEG Frequency Spectrum')
    plt.xlabel('Frequency (Hz)')
    plt.ylabel('Magnitude')
    plt.show()

# Main function
def main():
    print("Collecting EEG data...")
    eeg_data = read_eeg_data(ser, samples_to_collect)

    # Assuming a sampling rate typical for EEG data, e.g., 512 Hz
    sampling_rate = 512

    # Perform FFT on the collected data
    fft_freq, fft_magnitude = perform_fft(eeg_data, sampling_rate)

    # Plot the frequency spectrum
    plot_fft(fft_freq, fft_magnitude)
```

EEG Dataset From TGAM Module To Excel

	A	B	C	D	E	F
1	Timestamp	EEG_Freaquency	Alpha	Beta	Gamma	Attention
2	22/07/2024 21:21	17.92857143	0.8825944	0.98451793	0.6301169	77.5
3	22/07/2024 21:21	19.72215682	0.1079433	0.7911229	0.1934045	77.5
4	22/07/2024 21:21	2	0.6708877	0.90555989	0.85991586	5
5	22/07/2024 21:21	19.43530997	0.3958426	0.97428977	0.8086634	77.5
6	22/07/2024 21:21	19.16576577	0.1375393	0.32732315	0.76606036	77.5
7	22/07/2024 21:21	15.88502994	0.8987108	0.69871959	0.30585034	77.5
8	22/07/2024 21:21	18.56826568	0.5842285	0.84554922	0.29958185	77.5
9	22/07/2024 21:21	21.18317266	0.4174258	0.92072355	0.65851339	92.5
10	22/07/2024 21:21	19.89549703	0.9837028	0.72240853	0.76573522	77.5
11	22/07/2024 21:21	18.07885825	0.7769288	0.35148261	0.87136179	77.5
12	22/07/2024 21:21	14.5	0.9668807	0.02962936	0.7900321	60
13	22/07/2024 21:21	22.159583	0.3965655	0.51447588	0.43917472	92.5
14	22/07/2024 21:21	21.20711111	0.6493675	0.57604117	0.60804182	92.5
15	22/07/2024 21:21	19.69983949	0.601124	0.56822587	0.99019045	77.5
16	22/07/2024 21:21	8.1	0.264541	0.32010623	0.97484283	37.5
17	22/07/2024 21:21	21.38932889	0.7977055	0.36106616	0.15869727	92.5
18	22/07/2024 21:21	18.71303075	0.6009841	0.47851443	0.91133649	77.5
19	22/07/2024 21:21	19.07899807	0.7543736	0.32658556	0.59004864	77.5
20	22/07/2024 21:21	21.55457464	0.4247999	0.06829837	0.46692712	92.5
21	22/07/2024 21:21	21.01593625	0.0991563	0.47044364	0.17266834	92.5

1936	18/08/2024 0:43	9.681715576	0.5920612	0.09974232	0.16268706	37.5
1937	18/08/2024 0:43	11.33858268	0.3189144	0.08549166	0.87449485	37.5
1938	18/08/2024 0:43	10.52	0.4451378	0.92947829	0.26902076	37.5
1939	18/08/2024 0:43	8.322097378	0.6354166	0.67154087	0.28479191	37.5
1940	18/08/2024 0:43	8.849941383	0.6885649	0.39312865	0.16283028	37.5
1941	18/08/2024 0:43	9.718213058	0.0929118	0.35033016	0.88101581	37.5
1942	18/08/2024 0:43	8.469512195	0.4697206	0.55746343	0.7086731	37.5
1943	18/08/2024 0:43	7.350877193	0.2867571	0.56951226	0.06698804	20
1944	18/08/2024 0:43	9.391791045	0.9157221	0.02946032	0.12248605	37.5
1945	18/08/2024 0:43	8.094405594	0.2809788	0.6362471	0.96859589	37.5
1946	18/08/2024 0:43	8.442028986	0.0641208	0.91687637	0.91593631	37.5
1947	18/08/2024 0:43	9.523809524	0.5506347	0.93913933	0.40887754	37.5
1948	18/08/2024 0:43	5.997716895	0.5630688	0.34350322	0.39593772	20
1949	18/08/2024 0:43	7.990390775	0.3790827	0.67116297	0.37762808	20
1950	18/08/2024 0:43	12.38888889	0.2134714	0.892023	0.78316332	60
1951	18/08/2024 1:25	24.64838337	0.9966391	0.18571871	0.47560557	92.5
1952	18/08/2024 1:25	24.67708333	0.454394	0.68577073	0.64466321	92.5
1953	18/08/2024 1:25	24.62366442	0.324803	0.41405058	0.21074923	92.5
1954	18/08/2024 1:25	24.69969653	0.2323063	0.11791958	0.97155978	92.5
1955	18/08/2024 20:57	23.9945322	0.8125394	0.49139768	0.38636325	92.5
1956	18/08/2024 20:57	25.34635008	0.5534038	0.29008116	0.35809523	92.5

DATA SET from KAGGLE:-

9	0.001978
11	0.002417
17	0.003735
5	0.001099
9	0.001978
14	0.003076
23	0.005054
42	0.009229
58	0.012744
55	0.012085
36	0.00791
6	0.001318
6	0.001318
18	0.003955
44	0.009668
60	0.013184
43	0.009448
6	0.001318
9	0.001978
1	0.00022
24	0.005273

Raw Data	ÊµÊ±µçÑ¹	m
66	0.014502	
50	0.010986	
41	0.009009	
43	0.009448	
49	0.010767	
44	0.009668	
28	0.006152	
8	0.001758	
21	0.004614	
37	0.00813	
18	0.003955	
2	0.000439	
23	0.005054	
37	0.00813	
29	0.006372	
23	0.005054	
37	0.00813	
66	0.014502	
81	0.017798	
102	0.022412	

A	B
29	0.006372
23	0.005054
37	0.00813
66	0.014502
81	0.017798
102	0.022412
128	0.028125
140	0.030762
144	0.031641
135	0.029663
134	0.029443
152	0.033398
165	0.036255
172	0.037793
178	0.039111
178	0.039111
173	0.038013
165	0.036255
162	0.035596
157	0.034497

9	0.001978
11	0.002417
17	0.003735
5	0.001099
9	0.001978
14	0.003076
23	0.005054
42	0.009229
58	0.012744
55	0.012085
36	0.00791
6	0.001318
6	0.001318
18	0.003955
44	0.009668
60	0.013184
43	0.009448
6	0.001318
9	0.001978
1	0.00022
24	0.005273

Experiment No # 1

Working on processing EEG (Electroencephalogram) signals and analyzing different frequency bands such as gamma, beta, alpha, theta, and delta. The code appears to be using a wavelet decomposition technique to extract these frequency components and perform spectral analysis.

MATLAB CODE:-

matlab Copy code

```
% Import EEG data
aa = importdata('nuresky_brain_data.csv');
eegdata = aa.data;
s = eegdata(2:end, 1);
```

- We import EEG data from a CSV file named 'nuresky_brain_data.csv' using the `importdata` function.
- The EEG data is extracted from the imported data structure and stored in the variable `s`.

matlab Copy code

```
% Plot the EEG signal
figure;
subplot(4,1,1);
plot(s);
title('EEG Signal');
```

- We create a new figure and add a subplot for the EEG signal.
- The EEG signal is plotted in the first subplot.
- We set the title of the subplot to 'EEG Signal'.

***Wavelet decomposition is a method used to break down a complex signal (like an EEG signal) into simpler parts, called wavelets, which are easier to analyze wavelet decomposition separates your signal into different frequency bands**

DATA SETS

	A	B	C	D	E	F	G	H
1	Timestamp	EEG_Frea	Alpha	Beta	Gamma	Attention		
2	22/07/2024 21:21	17.92857	0.882594	0.984518	0.630117	77.5		
3	22/07/2024 21:21	19.72216	0.107943	0.791123	0.193404	77.5		
4	22/07/2024 21:21	2	0.670888	0.90556	0.859916	5		
5	22/07/2024 21:21	19.43531	0.395843	0.97429	0.808663	77.5		
6	22/07/2024 21:21	19.16577	0.137539	0.327323	0.76606	77.5		
7	22/07/2024 21:21	15.88503	0.898711	0.69872	0.30585	77.5		
8	22/07/2024 21:21	18.56827	0.584229	0.845549	0.299582	77.5		
9	22/07/2024 21:21	21.18317	0.417426	0.920724	0.658513	92.5		
10	22/07/2024 21:21	19.8955	0.983703	0.722409	0.765735	77.5		
11	22/07/2024 21:21	18.07886	0.776929	0.351483	0.871362	77.5		
12	22/07/2024 21:21	14.5	0.966881	0.029629	0.790032	60		
13	22/07/2024 21:21	22.15958	0.396565	0.514476	0.439175	92.5		
14	22/07/2024 21:21	21.20711	0.649368	0.576041	0.608042	92.5		
15	22/07/2024 21:21	19.69984	0.601124	0.568226	0.99019	77.5		
16	22/07/2024 21:21	8.1	0.264541	0.320106	0.974843	37.5		
17	22/07/2024 21:21	21.38933	0.797705	0.361066	0.158697	92.5		
18	22/07/2024 21:21	18.71303	0.600984	0.478514	0.911336	77.5		
19	22/07/2024 21:21	19.079	0.754374	0.326586	0.590049	77.5		
20	22/07/2024 21:21	21.55457	0.4248	0.068298	0.466927	92.5		
21	22/07/2024 21:21	21.01594	0.099156	0.470444	0.172668	92.5		

```python
import pandas as pd

# Load the EEG data from the uploaded CSV file
file_path = '/mnt/data/concentration_data.csv'
eeg_data = pd.read_csv(file_path)

# Display the first few rows of the dataset to understand its structure
eeg_data.head()
```

Result

	Timestamp	EEG Freaquency	Alpha	Beta	Gamma	Atten
0	22/07/2024 21:21	17.928571	0.882594	0.984518	0.630117	
1	22/07/2024 21:21	19.722157	0.107943	0.791123	0.193404	
2	22/07/2024 21:21	2.000000	0.670888	0.905560	0.859916	
3	22/07/2024 21:21	19.435310	0.395843	0.974290	0.808663	
4	22/07/2024 21:21	19.165766	0.137539	0.327323	0.766060	

RESULTS:-

Editor - C:\Users\PMLS\Downloads\EEG\eegnew.m

eegnew.m

```
% Load the EEG data from the CSV file
data = readtable('concentration_data.csv');

% Extract relevant data columns
eeg_signal = data.EEG_Freaquency;
alpha_signal = data.Alpha;
beta_signal = data.Beta;
gamma_signal = data.Gamma;

% Time vector (assuming uniform sampling based on the length of the signal)
t = 1:length(eeg_signal);

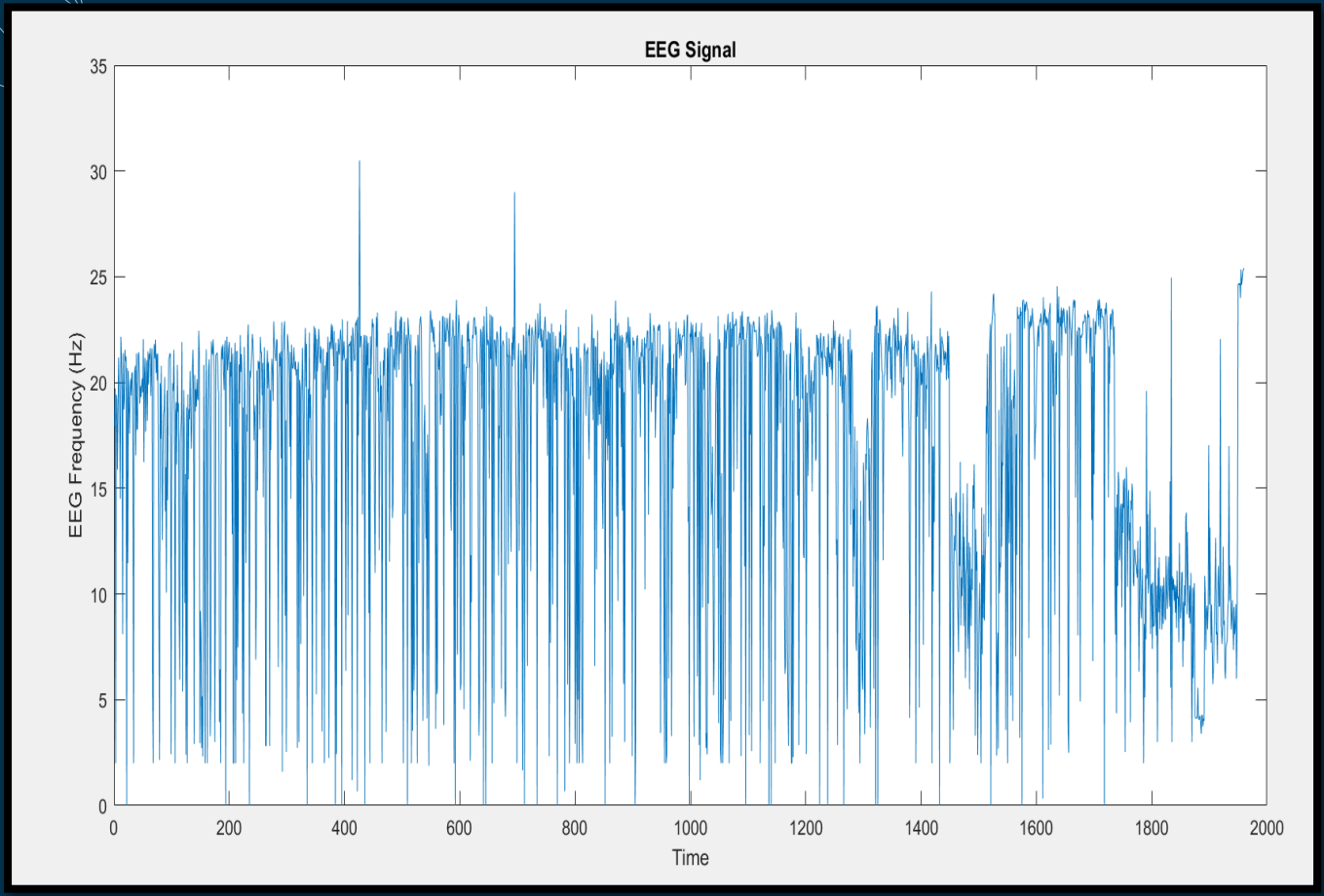
% Plot the original EEG signal in Figure 1
figure(1);
plot(t, eeg_signal);
title('EEG Signal');
```

```
>> eegnew
Average Alpha Frequency: 0.49
Average Beta Frequency: 0.50
Average Gamma Frequency: 0.49
```

- **Alpha Frequency (0.49):** Indicates that the alpha band (8-13 Hz) has a moderate presence in the EEG signal.
- **Beta Frequency (0.50):** Indicates that the beta band (13-30 Hz) has a slightly stronger presence compared to alpha and gamma.
- **Gamma Frequency (0.49):** Indicates that the gamma band (>30 Hz) has a moderate presence, similar to alpha.

***Normalize Formula:-** $\frac{Obatined - Minimmum}{Maximum - minimum}$

Graphical Representation

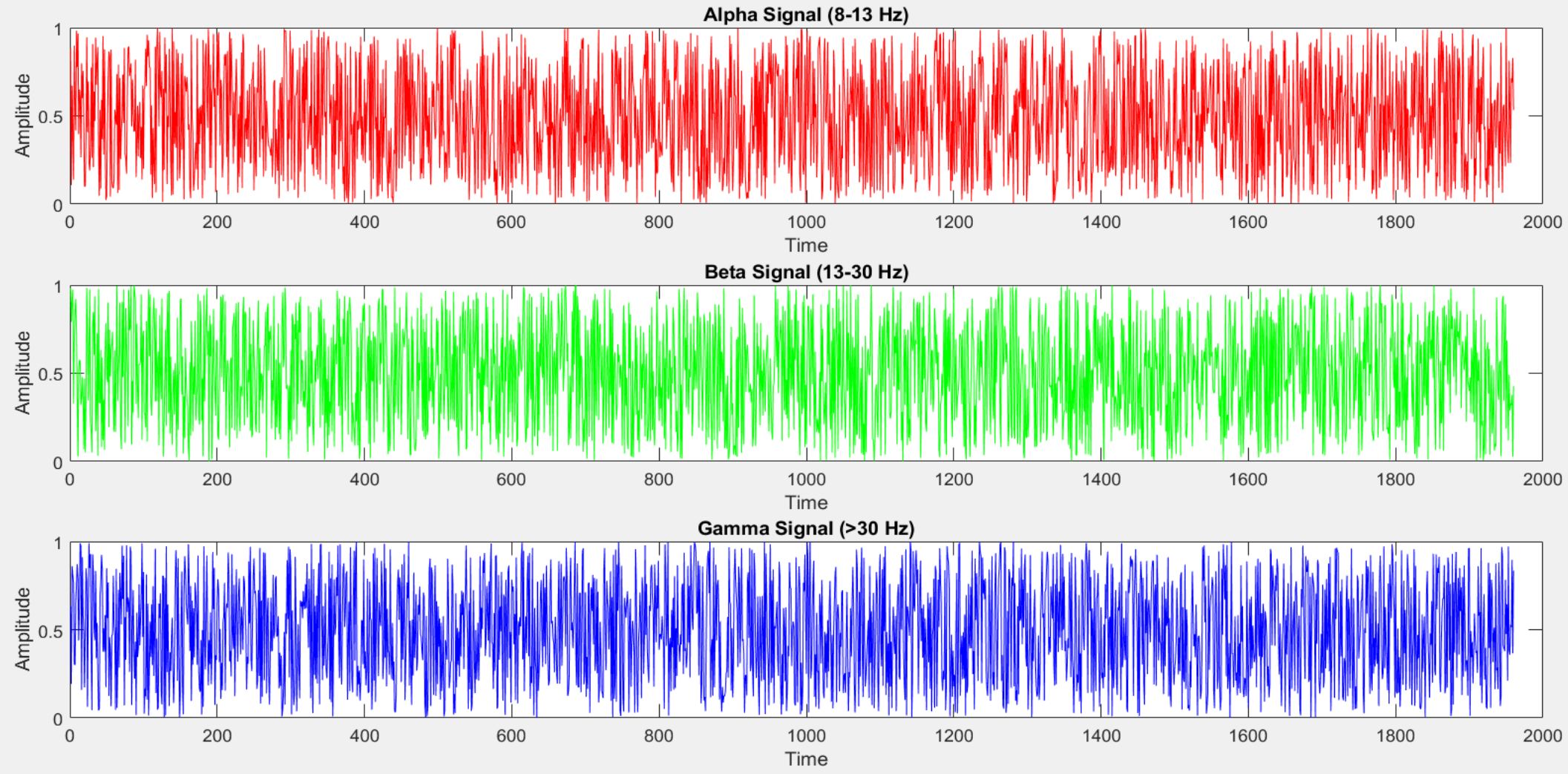


- **X-Axis (Time):** Represents the time points or sample indices during the recording.
- **Y-Axis (EEG Frequency (Hz)):** Represents the EEG signal's frequency component in Hertz (Hz).
- **Meaning:** This plot shows how the EEG frequency changes over time, reflecting overall brainwave activity.

- **X-axis ranges from 0 to 2000, and the sampling rate is 256 Hz:**

  2000 data points / 256 samples per second =approximately 7.8 seconds of recording.

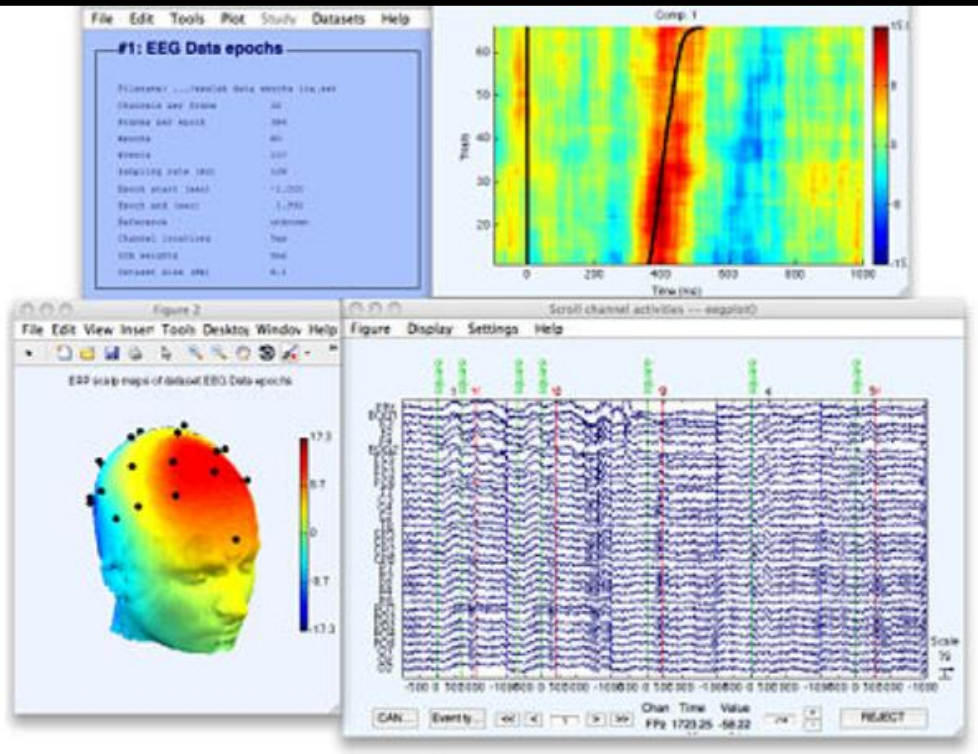
Graphical Representation



Experiment No # 2

EEGLAB:-

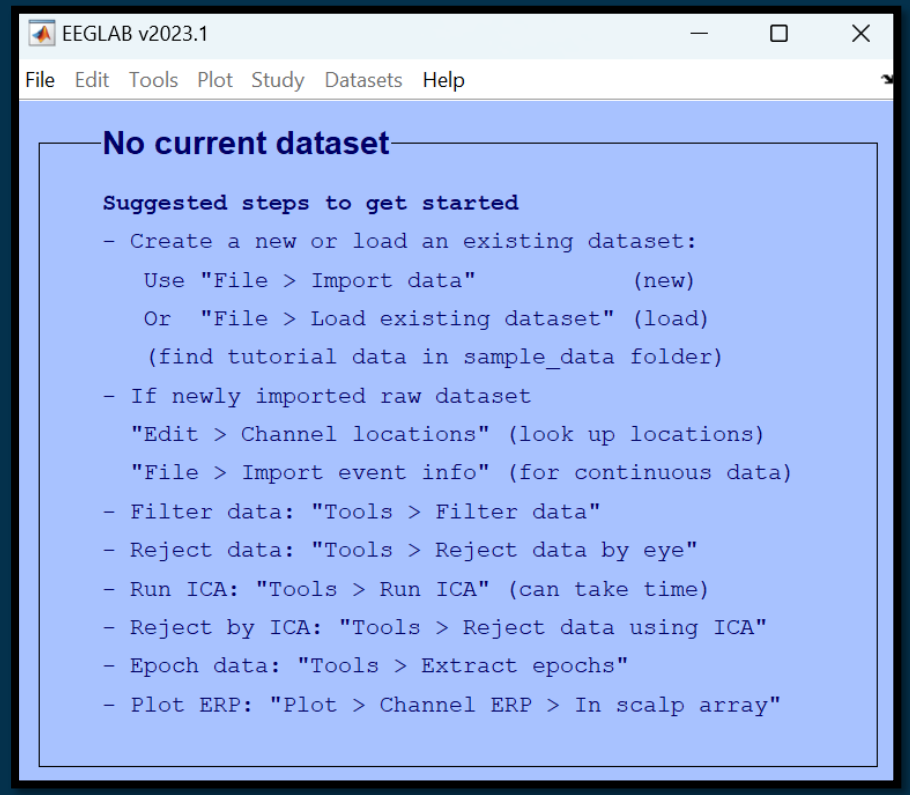
EEGLAB is an interactive Matlab toolbox for processing continuous and event-related EEG, MEG and other electrophysiological data incorporating independent component analysis (ICA), time/frequency analysis, artifact rejection, event-related statistics, and several useful modes of visualization of the averaged and single-trial data. EEGLAB runs under Linux, Unix, Windows, and Mac OS X.



EEGLAB provides an interactive graphic user interface (GUI) allowing users to flexibly and interactively process their high-density EEG and other dynamic brain data using independent component analysis (ICA) and/or time/frequency analysis (TFA), as well as standard averaging methods. EEGLAB

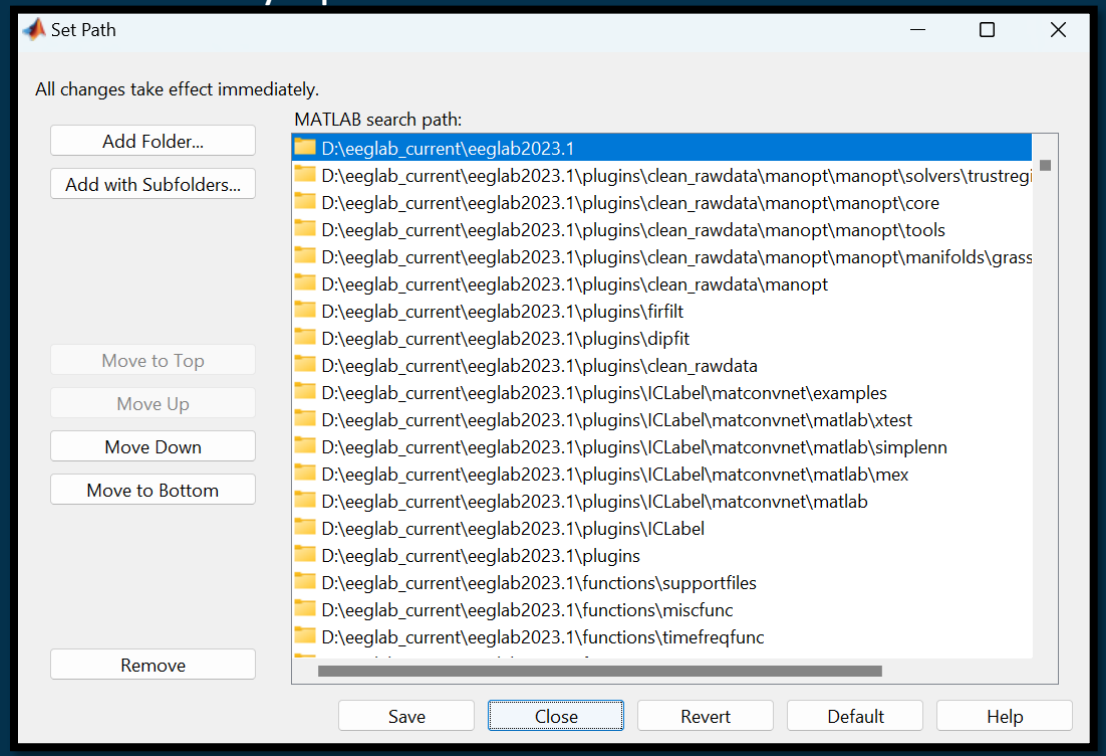
Start EEGLAB

Type eagle at the MATLAB command line and hit enter. EEGLAB will automatically add itself to the MATLAB path.

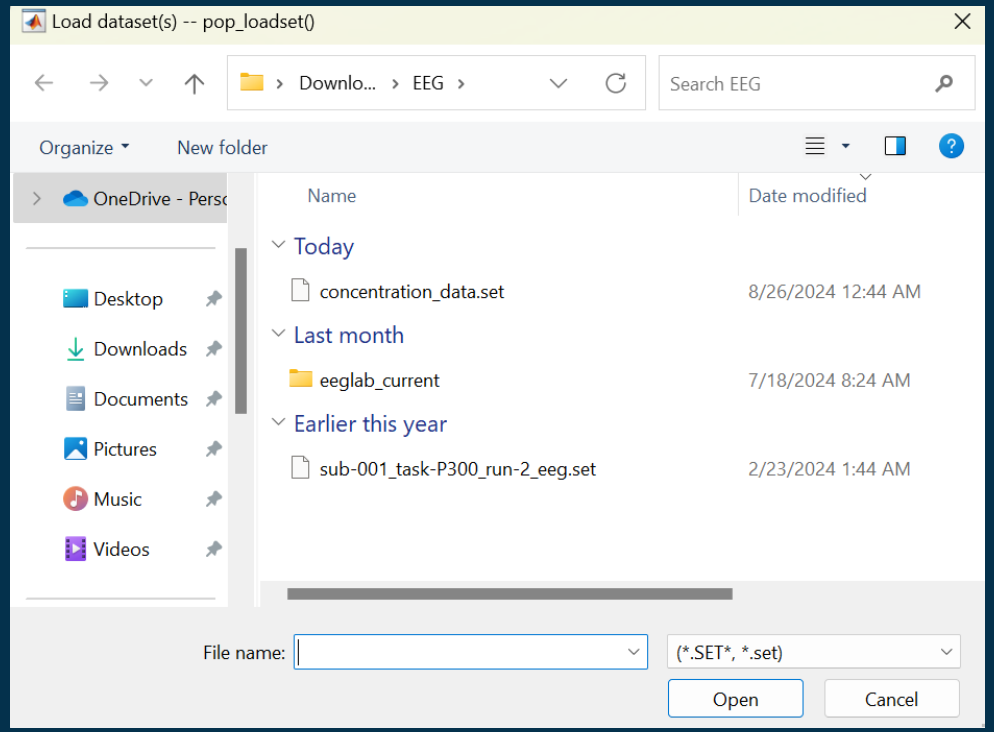


Adding EEGLAB to the MATLAB path

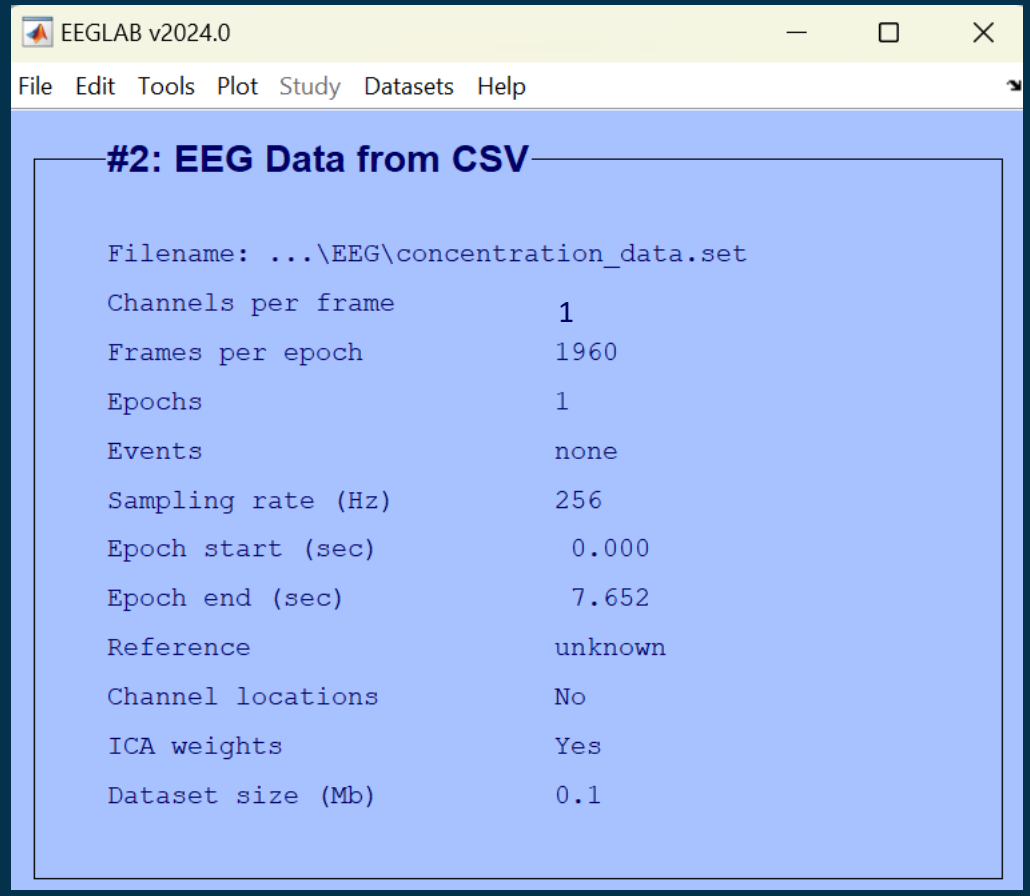
You may want to add the EEGLAB folder to the MATLAB search path so the next time you start MATLAB, you will be able to directly open EEGLAB.



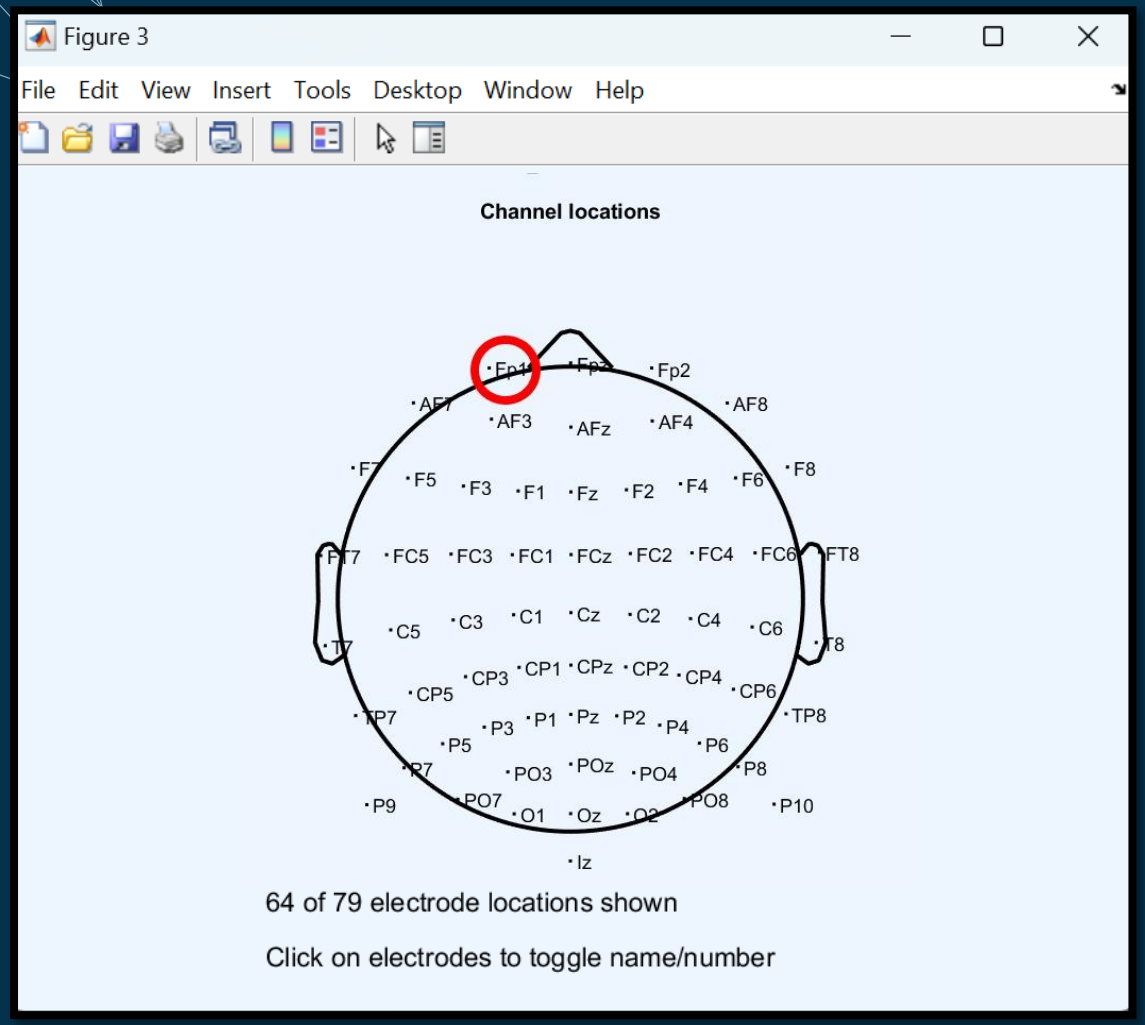
Load the sample EEGLAB dataset



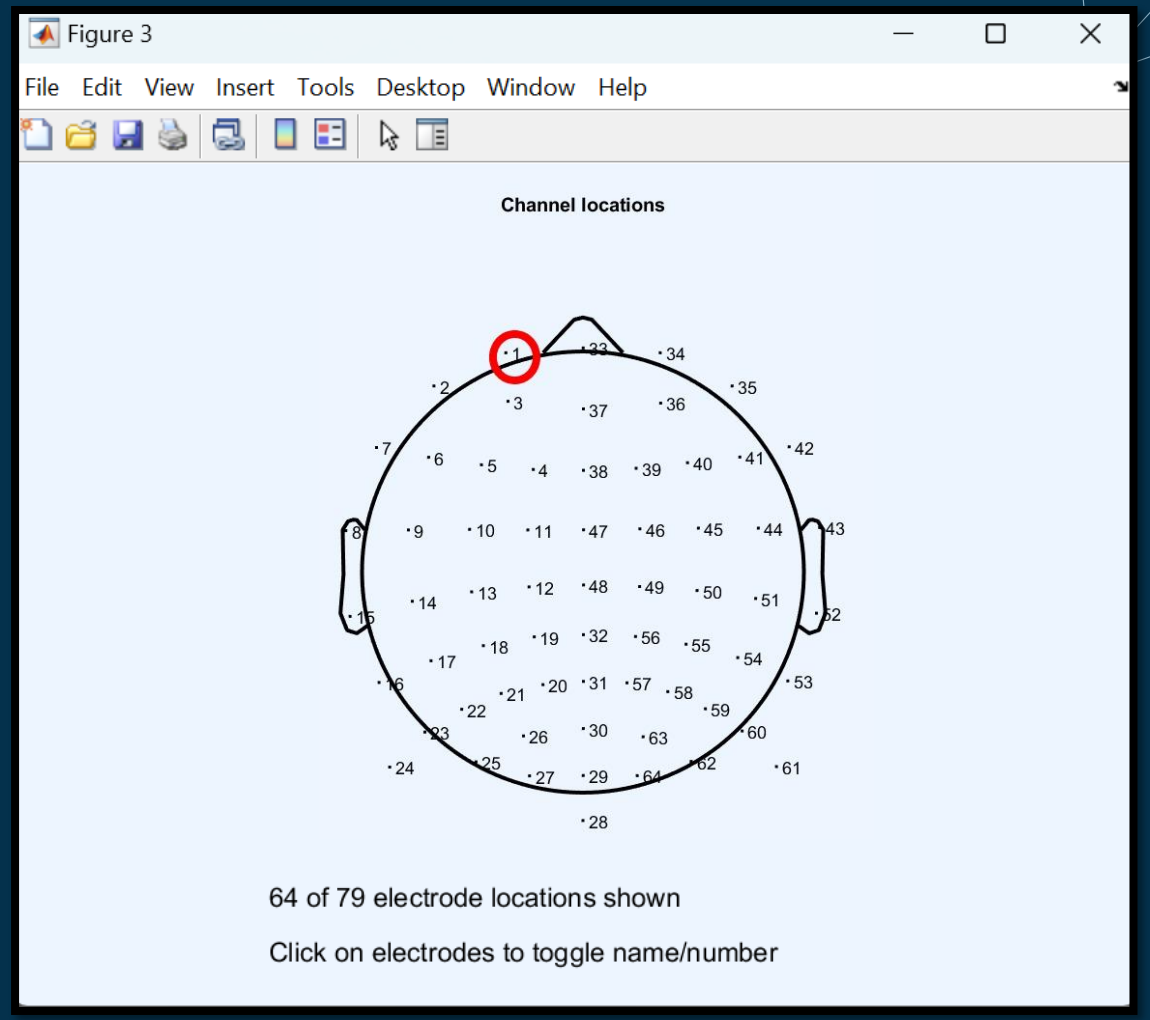
After EEGLAB loads the dataset, the main EEGLAB window shows relevant information about it – its number of channels, sampling rate, etc.



CHANNEL LOCATION BY NAME:

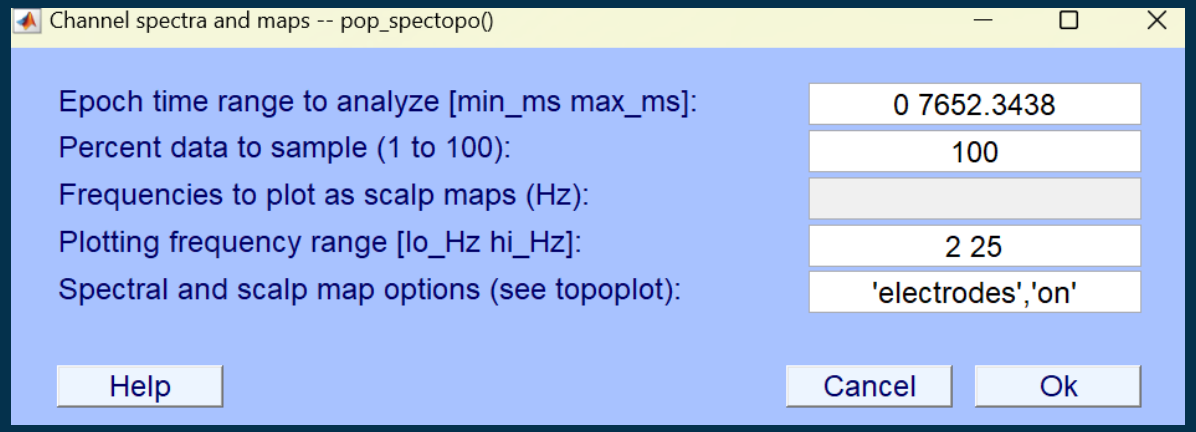


CHANNEL LOCATION BY NUMBER:



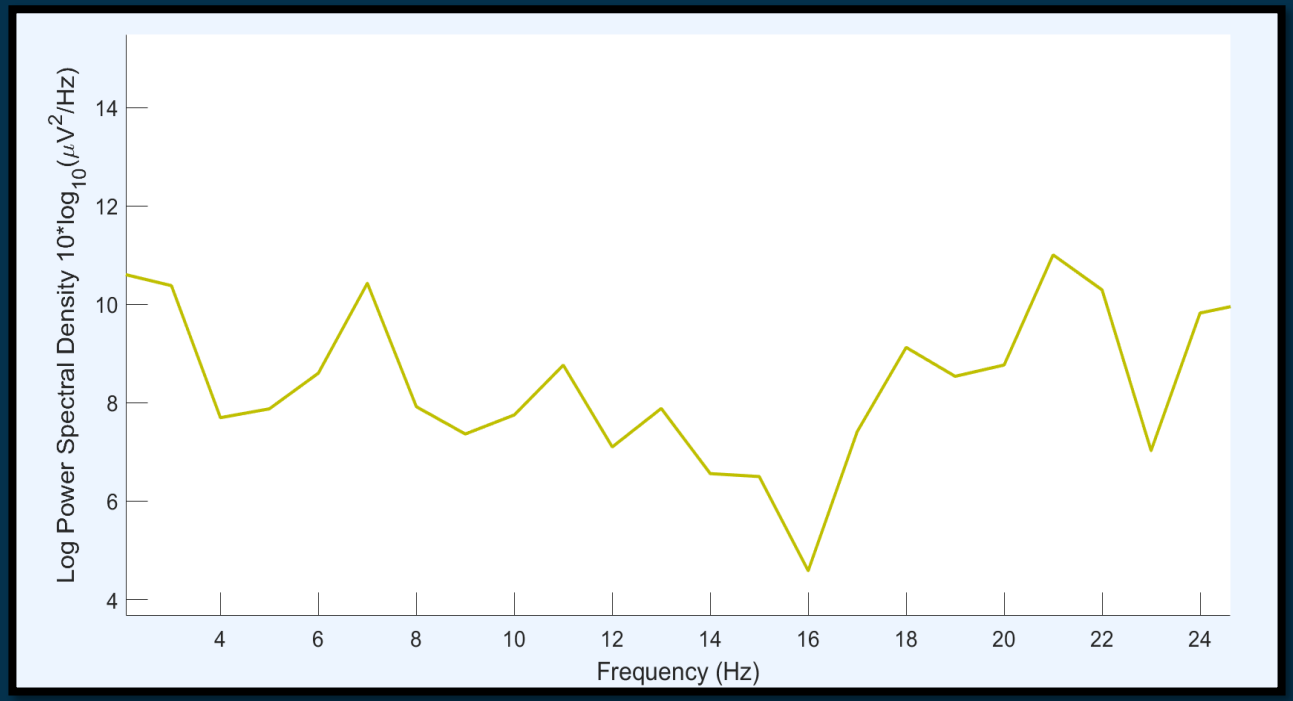
EEG Analysis

In the tutorial dataset, the EEG. Event structure fields type, position, and latency are specified for each of the events marked in the dataset.



Used for detecting brainwave patterns, cognitive states,

- Y-axis shows power (log scale) of brain activity;
- X-axis shows frequency (2-25 Hz).

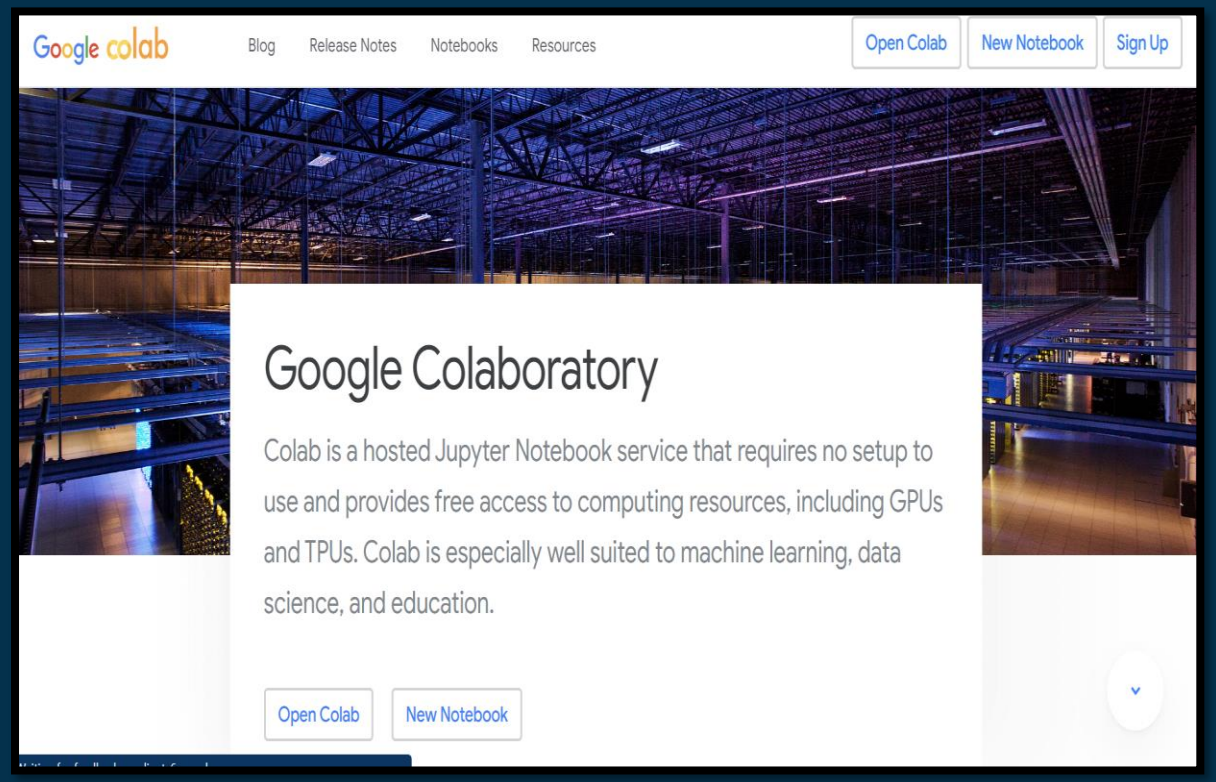


***Showing Frequency Power :**
Highest frequency More impact (observed around 7 Hz, 11 Hz, and 20 Hz)
Lowest Frequency less Impact (observed around 4 Hz, 10 Hz, and 16 Hz)

Experiment No # 3

AI AND MACHINE LEARNING:-

- Working with Python in Google Collab is a great choice for AI and machine learning projects, especially when processing EEG data
- EEG data from 2 Uni students who take class of 1 hours of same subject. The EEG data was recorded using a single-channel Mindset, measuring frontal lobe activity, with electrodes placed on the forehead and ears.



	A	B	C	D	E	F
1	Timestamp	EEG_Freaquency	Alpha	Beta	Gamma	Attention
2	22/07/2024 21:21	17.92857143	0.8825944	0.98451793	0.6301169	77.5
3	22/07/2024 21:21	19.72215682	0.1079433	0.7911229	0.1934045	77.5
4	22/07/2024 21:21	2	0.6708877	0.90555989	0.85991586	5
5	22/07/2024 21:21	19.43530997	0.3958426	0.97428977	0.8086634	77.5
6	22/07/2024 21:21	19.16576577	0.1375393	0.32732315	0.76606036	77.5
7	22/07/2024 21:21	15.88502994	0.8987108	0.69871959	0.30585034	77.5
8	22/07/2024 21:21	18.56826568	0.5842285	0.84554922	0.29958185	77.5
9	22/07/2024 21:21	21.18317266	0.4174258	0.92072355	0.65851339	92.5
10	22/07/2024 21:21	19.89549703	0.9837028	0.72240853	0.76573522	77.5
11	22/07/2024 21:21	18.07885825	0.7769288	0.35148261	0.87136179	77.5
12	22/07/2024 21:21	14.5	0.9668807	0.02962936	0.7900321	60
13	22/07/2024 21:21	22.159583	0.3965655	0.51447588	0.43917472	92.5
14	22/07/2024 21:21	21.20711111	0.6493675	0.57604117	0.60804182	92.5
15	22/07/2024 21:21	19.69983949	0.601124	0.56822587	0.99019045	77.5
16	22/07/2024 21:21	8.1	0.264541	0.32010623	0.97484283	37.5
17	22/07/2024 21:21	21.38932889	0.7977055	0.36106616	0.15869727	92.5
18	22/07/2024 21:21	18.71303075	0.6009841	0.47851443	0.91133649	77.5
19	22/07/2024 21:21	19.07899807	0.7543736	0.32658556	0.59004864	77.5
20	22/07/2024 21:21	21.55457464	0.4247999	0.06829837	0.46692712	92.5
21	22/07/2024 21:21	21.01593625	0.0991563	0.47044364	0.17266834	92.5

1936	18/08/2024 0:43	9.681715576	0.5920612	0.09974232	0.16268706	37.5
1937	18/08/2024 0:43	11.33858268	0.3189144	0.08549166	0.87449485	37.5
1938	18/08/2024 0:43	10.52	0.4451378	0.92947829	0.26902076	37.5
1939	18/08/2024 0:43	8.322097378	0.6354166	0.67154087	0.28479191	37.5
1940	18/08/2024 0:43	8.849941383	0.6885649	0.39312865	0.16283028	37.5
1941	18/08/2024 0:43	9.718213058	0.0929118	0.35033016	0.88101581	37.5
1942	18/08/2024 0:43	8.469512195	0.4697206	0.55746343	0.7086731	37.5
1943	18/08/2024 0:43	7.350877193	0.2867571	0.56951226	0.06698804	20
1944	18/08/2024 0:43	9.391791045	0.9157221	0.02946032	0.12248605	37.5
1945	18/08/2024 0:43	8.094405594	0.2809788	0.6362471	0.96859589	37.5
1946	18/08/2024 0:43	8.442028986	0.0641208	0.91687637	0.91593631	37.5
1947	18/08/2024 0:43	9.523809524	0.5506347	0.93913933	0.40887754	37.5
1948	18/08/2024 0:43	5.997716895	0.5630688	0.34350322	0.39593772	20
1949	18/08/2024 0:43	7.990390775	0.3790827	0.67116297	0.37762808	20
1950	18/08/2024 0:43	12.38888889	0.2134714	0.892023	0.78316332	60
1951	18/08/2024 1:25	24.64838337	0.9966391	0.18571871	0.47560557	92.5
1952	18/08/2024 1:25	24.67708333	0.454394	0.68577073	0.64466321	92.5
1953	18/08/2024 1:25	24.62366442	0.324803	0.41405058	0.21074923	92.5
1954	18/08/2024 1:25	24.69969653	0.2323063	0.11791958	0.97155978	92.5
1955	18/08/2024 20:57	23.9945322	0.8125394	0.49139768	0.38636325	92.5
1956	18/08/2024 20:57	25.34635008	0.5534038	0.29008116	0.35809523	92.5

Start Machine Learning

An environment for a machine learning project, which involves classifying EEG data into different categories based on brainwave frequencies: alpha, beta, and gamma.

C: > Users > mhass > OneDrive > Desktop > Asad FYP > EEG Data > attention_predictor_model.pkl.py > ...

```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error
import pickle

# Load the training data (make sure this path is correct)
training_data_path = 'c:/Users/mhass/OneDrive/Desktop/Asad FYP/EEG Data/training_data.csv'
training_data = pd.read_csv(training_data_path)

# Features (EEG wave data)
X = training_data[['Low_Alpha', 'High_Alpha', 'Low_Beta', 'High_Beta', 'Low_Gamma', 'Mid_Gamma']]
# Target (Attention)
y = training_data['Attention']

# Split data into training and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Initialize and train the model
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Test the model
y_pred = model.predict(X_test)
print('Mean Squared Error:', mean_squared_error(y_test, y_pred))

# Save the trained model to a file
model_save_path = 'c:/Users/mhass/OneDrive/Desktop/Asad FYP/EEG Data/attention_predictor_model.pkl'
with open(model_save_path, 'wb') as f:
    pickle.dump(model, f)
```

Different Libraries:-

The first cell imports various Python libraries necessary for data analysis, visualization, machine learning, and deep learning:

- `pandas` and `numpy` for data manipulation,
- `matplotlib.pyplot` and `seaborn` for data visualization,
- `tensorflow` for building neural network models,
- `sklearn` for data preprocessing and model evaluation tools,
- various `tensorflow.keras.layers` for building neural network layers,
- `tensorflow.keras.models` for compiling the full model structure,
- `tensorflow.keras.optimizers` for the optimization algorithm during training,
- `tensorflow.keras.utils` for additional utilities such as plotting the model,
- `sklearn.metrics` for evaluating the model performance, like generating a confusion matrix.

Machine Learning Algorithms:

- **Support Vector Machine (SVM):** Used for classifying different mental states (e.g., attention vs mediation) based on the extracted features.
- **Convolutional Neural Networks (CNNs):** Applied for pattern recognition in the EEG data, particularly for identifying complex brainwave patterns associated with learning states.
- **Recurrent Neural Networks (RNNs) with LSTM:** These are used to capture the temporal dependencies in the EEG signals, predicting student engagement over time based on historical data.

Algorithm

```
def read_tgam_data(serial_port):
    ser = serial.Serial(serial_port, 57600)  # Open serial port
    while True:
        if ser.read() == b'\xaa' and ser.read() == b'\xaa':  # Sync bytes
            packet_length = ser.read(1)[0]
            payload = ser.read(packet_length)
            checksum = ser.read(1)[0]
            if checksum == (0xFF - (sum(payload) & 0xFF)):
                attention, high_alpha, high_beta, mid_gamma = parse_payload(pay
                if attention is not None and high_alpha is not None:
                    update_data(attention, high_alpha, high_beta, mid_gamma)
                    save_attention_data(attention)
                    update_gui(attention, high_alpha, high_beta, mid_gamma)
```

DATA READ:

1960 rows × 6 columns

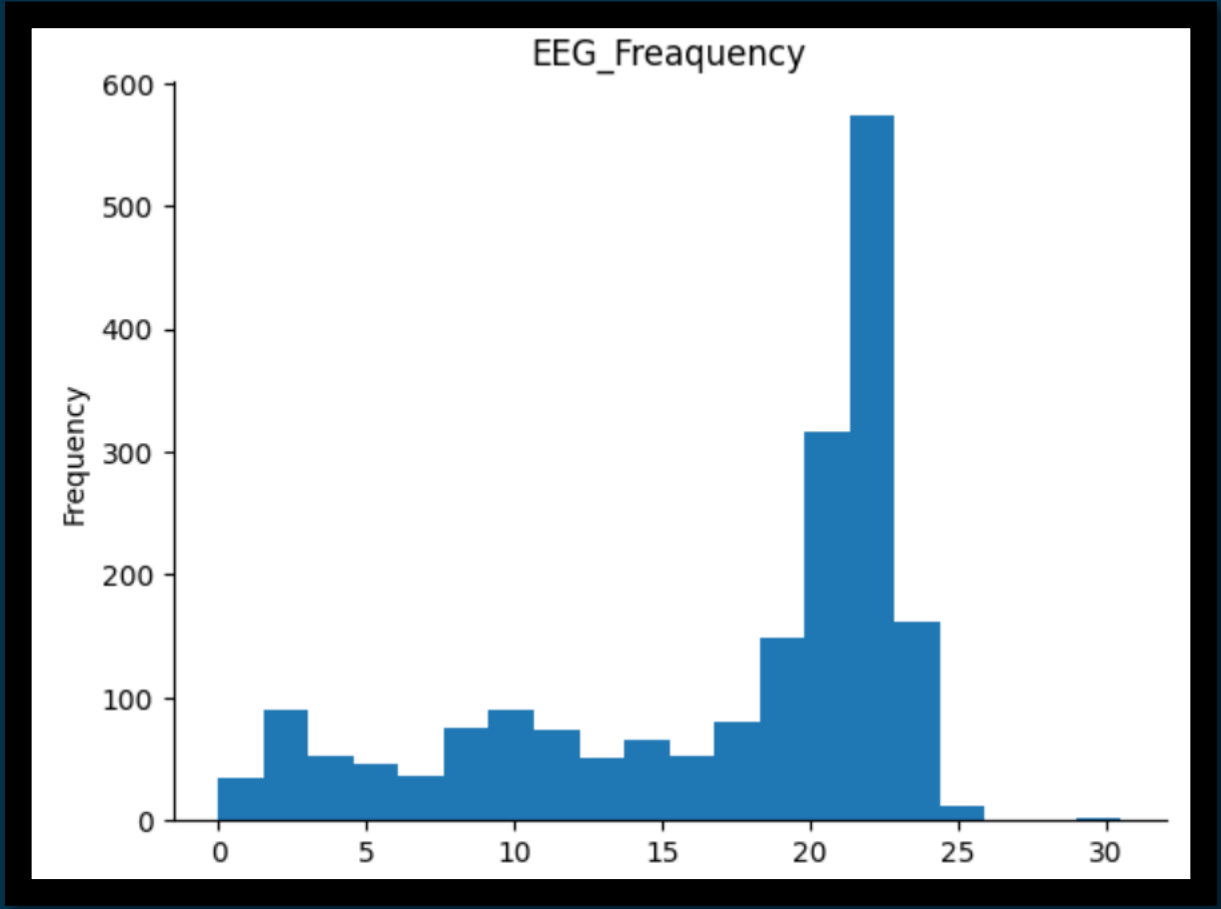
	Timestamp	EEG_Freaquency	Alpha	Beta	Gamma	Attention
0	22/07/2024 21:21	17.928571	0.882594	0.984518	0.630117	77.5
1	22/07/2024 21:21	19.722157	0.107943	0.791123	0.193404	77.5
2	22/07/2024 21:21	2.000000	0.670888	0.905560	0.859916	5.0
3	22/07/2024 21:21	19.435310	0.395843	0.974290	0.808663	77.5
4	22/07/2024 21:21	19.165766	0.137539	0.327323	0.766060	77.5
...	...	...	...	...	...	...
1955	18/08/2024 20:57	24.618890	0.233608	0.327996	0.528642	92.5
1956	18/08/2024 20:57	24.832794	0.713052	0.319069	0.513781	92.5
1957	18/08/2024 20:57	25.200334	0.687771	0.377180	0.895643	92.5
1958	18/08/2024 20:57	25.350186	0.828335	0.028009	0.366251	92.5
1959	18/08/2024 20:57	25.406714	0.532781	0.427385	0.834617	92.5

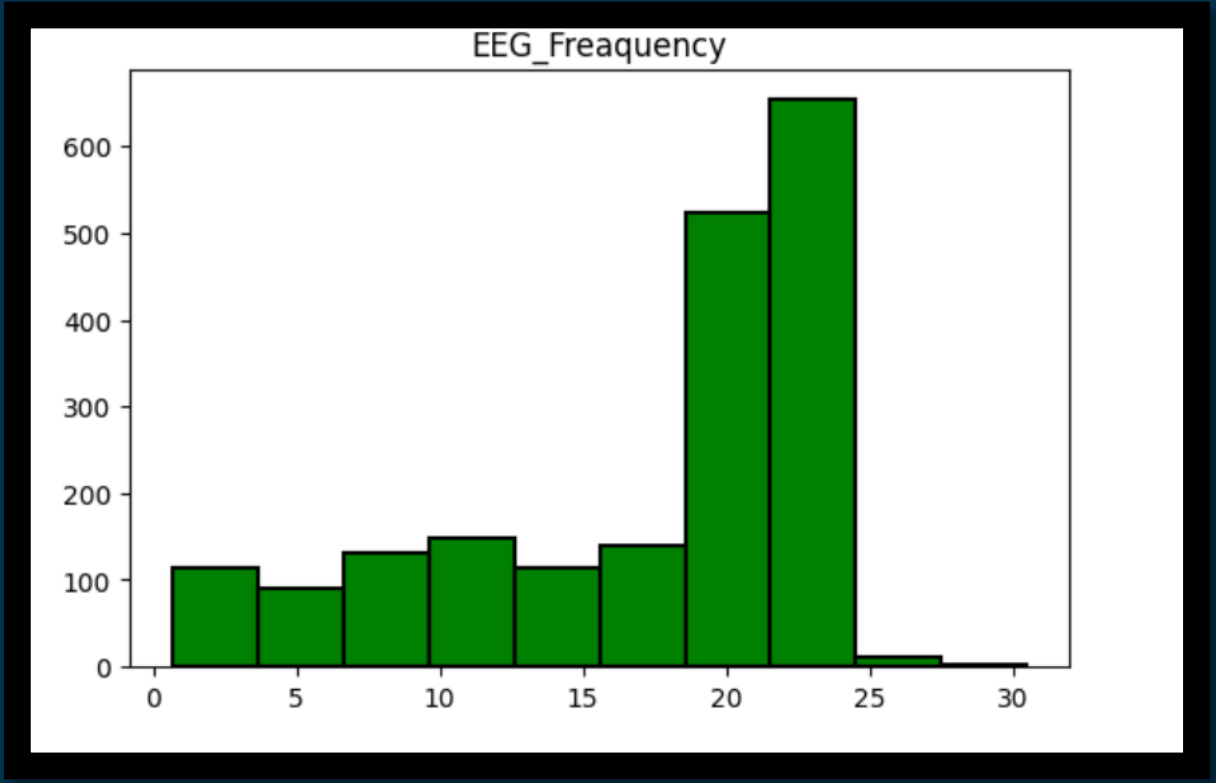
Each Student Status:
summary table of unique value counts for different variables in your EEG dataset.

	count of unique values
Timestamp	58
EEG_Freaquency	1856
Alpha	1960
Beta	1960
Gamma	1960

Histogram

The x-axis represents the count of unique values within each EEG feature, and the y-axis represents the frequency of EEG features within those unique value ranges.





This is a good sign for data quality, as it means you can proceed with analysis without needing to handle missing data.

```
data.isnull().sum()
```

	0
Timestamp	0
EEG_Freaquency	0
Alpha	0
Beta	0
Gamma	0
Attention	0

dtype: int64

- **count: The number of non-missing values in each column.**
- **mean: The average of the values.**
- **std: The standard deviation, a measure of the dispersion of the dataset.**
- **min: The smallest value.**
- **25%: The 25th percentile (first quartile), where 25% of the data points are below this value.**
- **50%: The 50th percentile (median), where half of the data is below this value.**
- **75%: The 75th percentile (third quartile), where 75% of the data points are below this value.**
- **max: The largest value.**

```
data.describe()
```

	EEG_Freaquency	Alpha	Beta	Gamma	Attention
count	1960.000000	1960.000000	1960.000000	1960.000000	1960.000000
mean	16.990129	0.490373	0.500538	0.493692	70.718112
std	6.750094	0.289213	0.286923	0.289069	29.867907
min	0.000000	0.000504	0.000955	0.000132	0.000000
25%	12.035354	0.235760	0.256145	0.236201	60.000000
50%	20.372559	0.481423	0.504330	0.490010	92.500000
75%	21.991160	0.737890	0.745560	0.748374	92.500000
max	30.500000	0.999261	0.999913	0.999997	100.000000

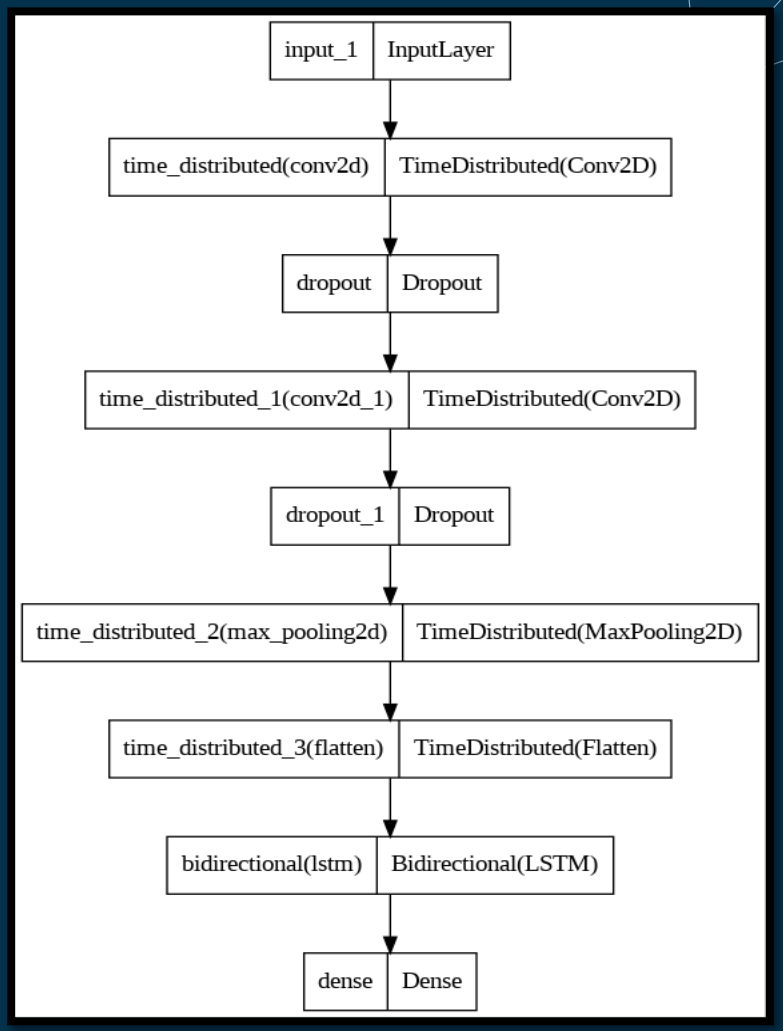
DATA Training:-

From training a machine learning model, specifically an LSTM (Long Short-Term Memory) neural network.

Architecture starts by extracting spatial features from the EEG data at each time step using convolutional layers.

```
#For dataset D1
Xtrain, Xtest, ytrain, ytest = train_test_split(D1, T, test_size=0.2, random_state=0)
r,model = lstm_model(Xtrain,Xtest,ytrain,ytest,Amin,11,epochs = 100)
```

```
_________________________________________________________________
Total params: 2093868 (7.99 MB)
Trainable params: 2093868 (7.99 MB)
Non-trainable params: 0 (0.00 Byte)
_________________________________________________________________
Epoch 1/100
3/3 - 6s - loss: 0.7027 - accuracy: 0.5750 - val_loss: 0.6041 - val_accuracy: 0.6500 - 6s/epoch - 2s/step
Epoch 2/100
3/3 - 0s - loss: 0.5878 - accuracy: 0.8500 - val_loss: 0.5818 - val_accuracy: 0.7500 - 303ms/epoch - 101ms/step
Epoch 3/100
3/3 - 0s - loss: 0.5588 - accuracy: 0.7750 - val_loss: 0.5130 - val_accuracy: 0.8000 - 285ms/epoch - 95ms/step
Epoch 4/100
3/3 - 0s - loss: 0.4691 - accuracy: 0.8125 - val_loss: 0.5723 - val_accuracy: 0.8000 - 271ms/epoch - 90ms/step
Epoch 5/100
3/3 - 0s - loss: 0.4157 - accuracy: 0.8250 - val_loss: 0.4891 - val_accuracy: 0.8000 - 268ms/epoch - 89ms/step
Epoch 6/100
3/3 - 0s - loss: 0.3586 - accuracy: 0.8375 - val_loss: 0.4874 - val_accuracy: 0.7500 - 294ms/epoch - 98ms/step
Epoch 7/100
3/3 - 0s - loss: 0.2744 - accuracy: 0.8875 - val_loss: 0.6407 - val_accuracy: 0.7000 - 270ms/epoch - 90ms/step
```



LSTM Training And Testing Data Flow For EEG

1.Input Layer: Takes in the EEG data as input.

2.Time Distributed Conv2D Layers: These layers look at each part of the EEG data over time to find important patterns.

3.Dropout Layers: Helps prevent the model from relying too much on any one part by randomly ignoring some parts during training.

4.Time Distributed MaxPooling2D: Simplifies the data by keeping only the most important information.

5.Time Distributed Flatten: Turns the data into a simple list of numbers so the model can process it more easily.

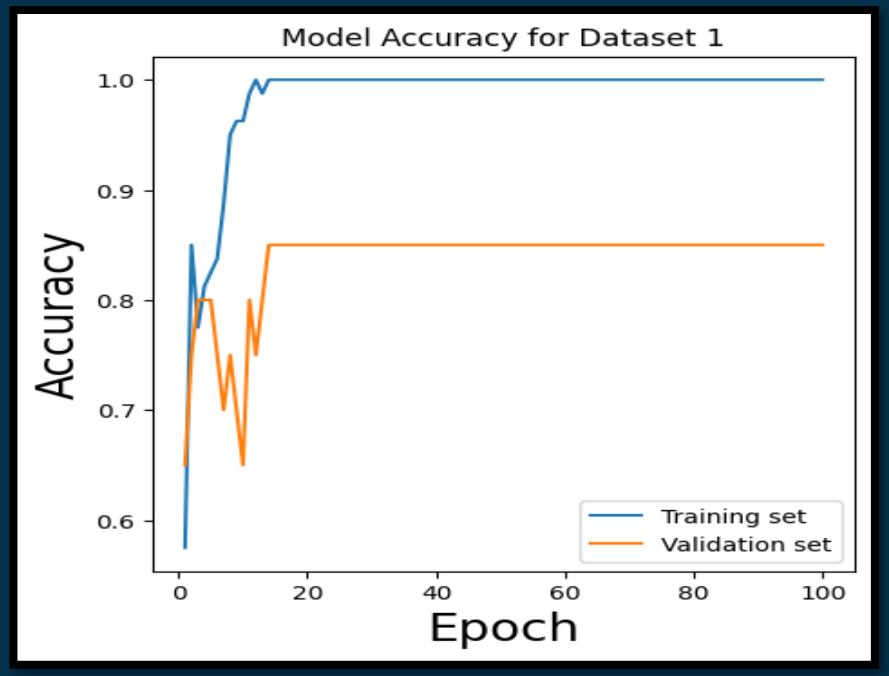
6.Bidirectional LSTM: Understands the sequence of the data by looking both forward and backward in time.

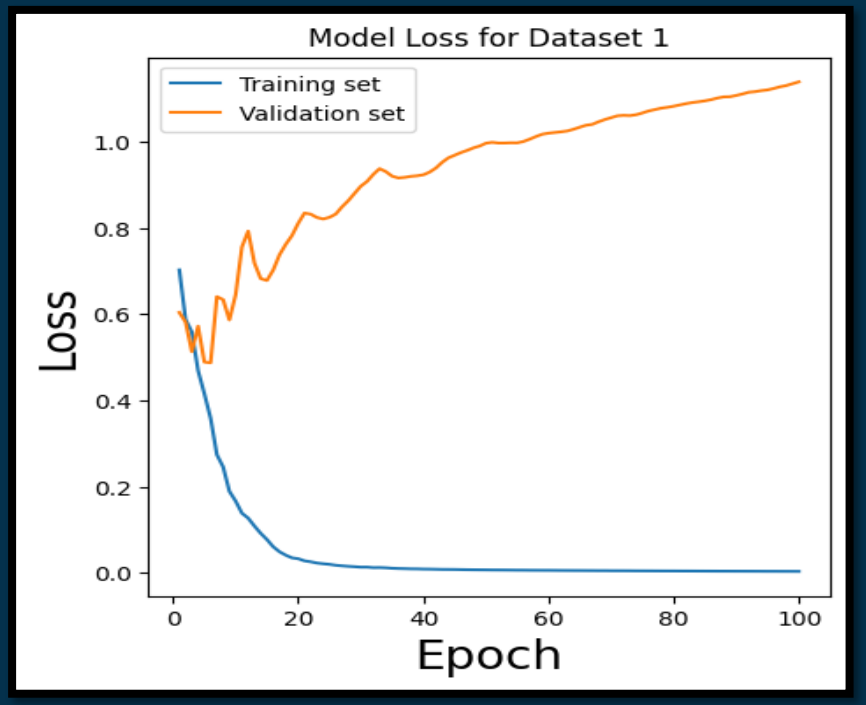
7.Dense Layer: Makes the final prediction or output based on everything the model has learned.

DATA TESTING:

The blue line represents the accuracy of the model on the training set.

The orange line represents the accuracy on the validation set. T





- **Training Accuracy (Blue):** Quickly reaches near 1.0, indicating strong learning on training data.
- **Validation Accuracy (Orange):** Stabilizes around 0.8, showing limited improvement after early epochs.
- **Overfitting Indication:** Large gap between training and validation accuracy suggests overfitting.
- **Early Stopping:** Could stop training around 20 epochs to avoid overfitting

- **Training Loss (Blue):** Decreases sharply to near 0, indicating the model is fitting well to the training data.
- **Validation Loss (Orange):** Increases after initial decrease, indicating overfitting as the model struggles to generalize.
- **Overfitting Sign:** Diverging trends between training and validation loss after early epochs.
- **Early Stopping:** Suggested to prevent further overfitting and high validation loss.

System Testing

Experiment No # 4

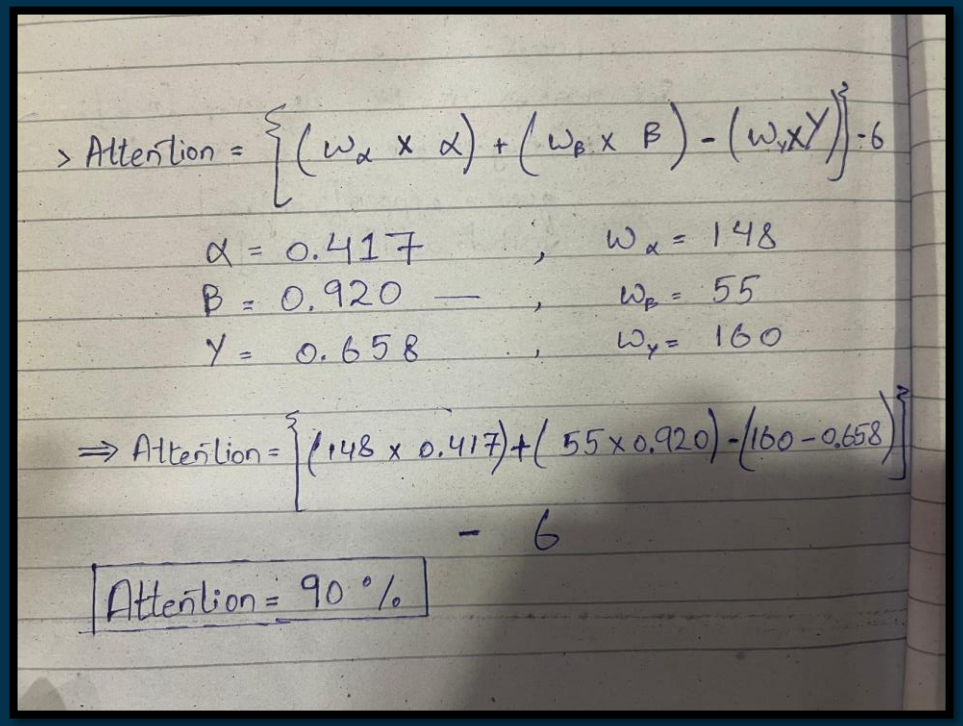
EEG Frequency Bands and Attention in Percentage

Brainwave Type	Frequency Range	Associated State	Approximate Attention Level (%)
Delta Waves	0.5 - 4 Hz	Deep sleep, unconsciousness	0% - 10%
Theta Waves	4 - 8 Hz	Drowsiness, light sleep, relaxation	10% - 25%
Alpha Waves	8 - 12 Hz	Relaxed, calm, but alert	25% - 50%
Low Beta Waves	12 - 15 Hz	Relaxed, calm, but alert	50% - 70%
Mid Beta Waves	15 - 20 Hz	Focused, alert, active thinking	70% - 85%
High Beta Waves	20 - 30 Hz	Intense focus, anxiety	85% - 100%

System Testing

Experiment No # 3

Formula for attention using linear Regression

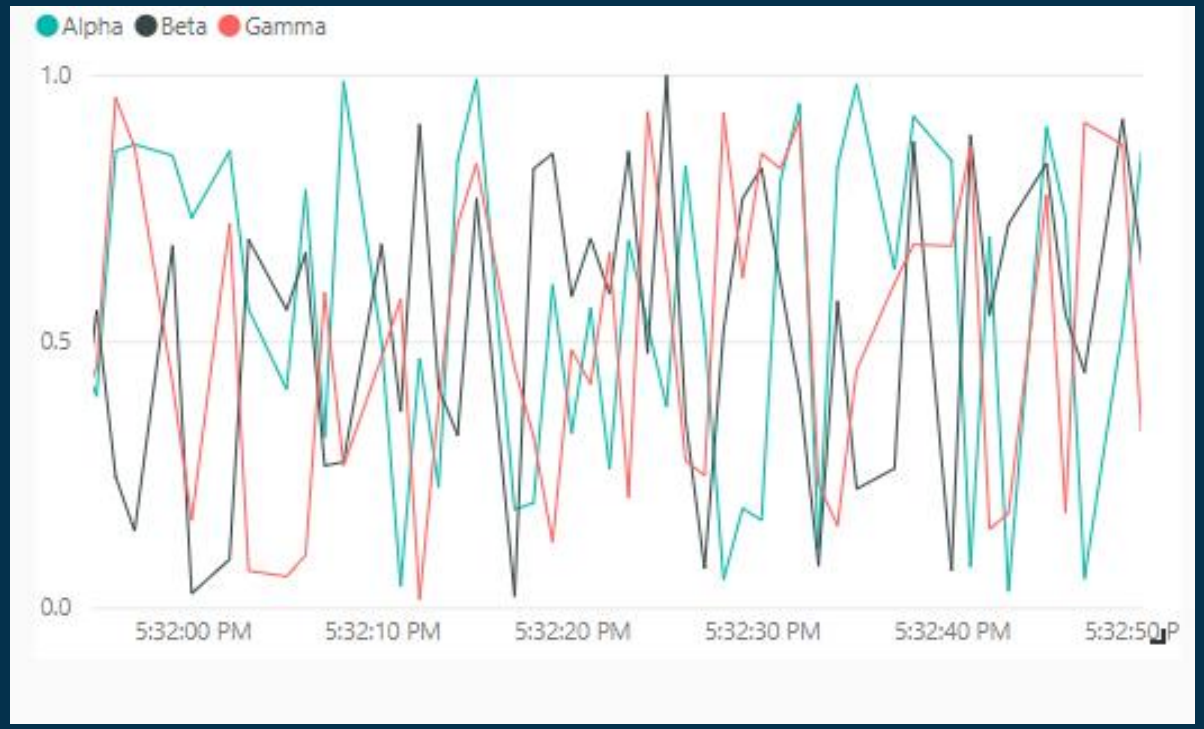


If the formula is linear, you can apply a multiple linear regression model where Attention is the dependent variable, and Alpha, Beta, and Gamma are the independent variables.

A	B	C	D	E	F
Timestam	EEG_Freac	Alpha	Beta	Gamma	Attention
22/07/202	17.92857	0.882594	0.984518	0.630117	77.5
22/07/202	19.72216	0.107943	0.791123	0.193404	77.5
22/07/202	2	0.670888	0.90556	0.859916	5
22/07/202	19.43531	0.395843	0.97429	0.808663	77.5
22/07/202	19.16577	0.137539	0.327323	0.76606	77.5
22/07/202	15.88503	0.898711	0.69872	0.30585	77.5
22/07/202	18.56827	0.584229	0.845549	0.299582	77.5
22/07/202	21.18317	0.417426	0.920724	0.658513	90

RESULTS-

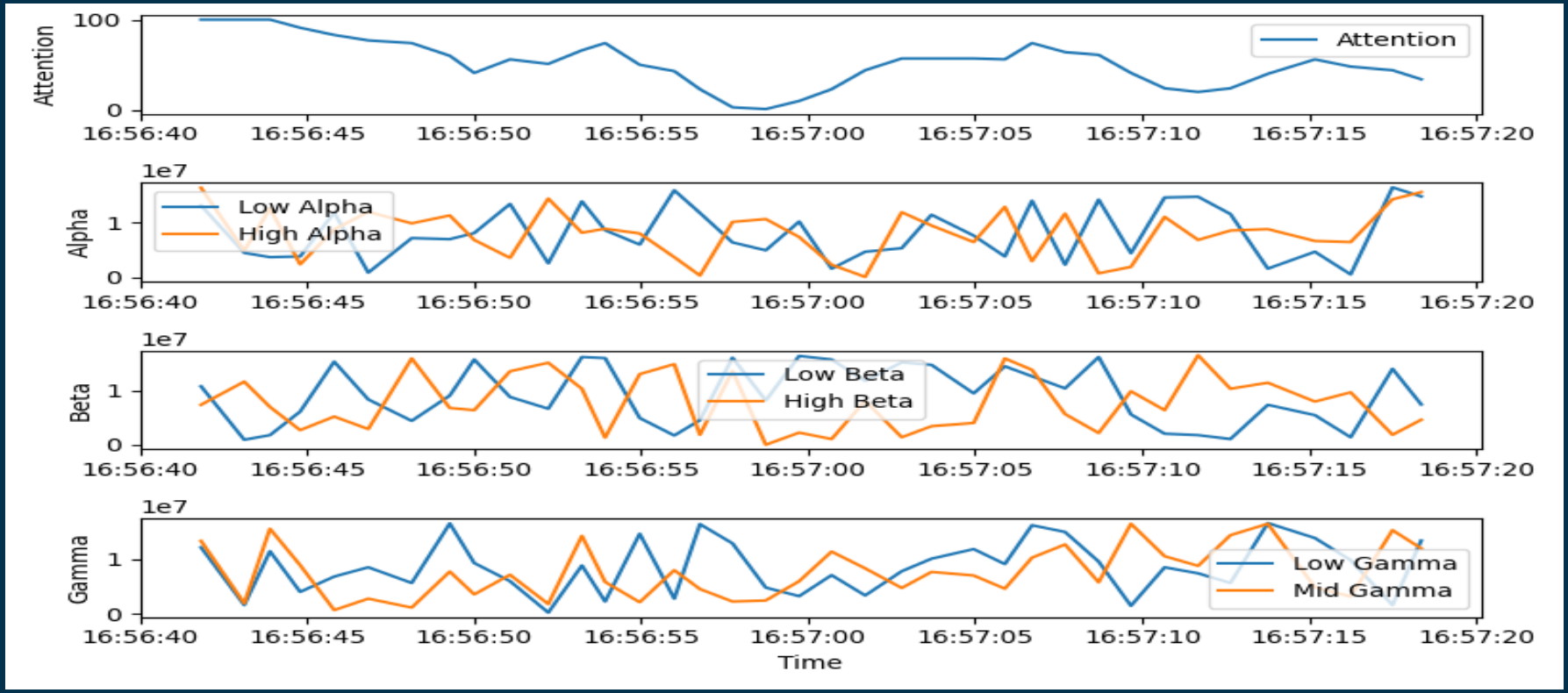
REALTIME EEG Alpha Beta Gamma Waves USING PYTHON



The first graph in the EEG data monitoring visualization of the state is the real-time Alpha, Beta and Gamma oscillations. The Y-axis gives the amplitude of these waves as the X-axis shows the time in seconds. Each wave type is color-coded for clarity: Alpha waves are represented by green, Beta waves by black while Gamma waves are represented by red

RESULTS-

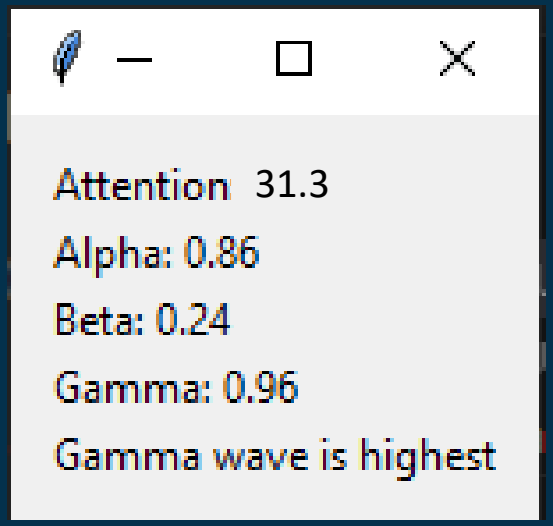
EEG Data Monitoring Visualization



This figure 1 provides a comprehensive overview of EEG data monitoring visualization provides a comprehensive overview of brain activity, segmented into four primary plots: Alertness, illustration, Brain Waves, Alpha Waves, Beta Waves, and Gamma Waves

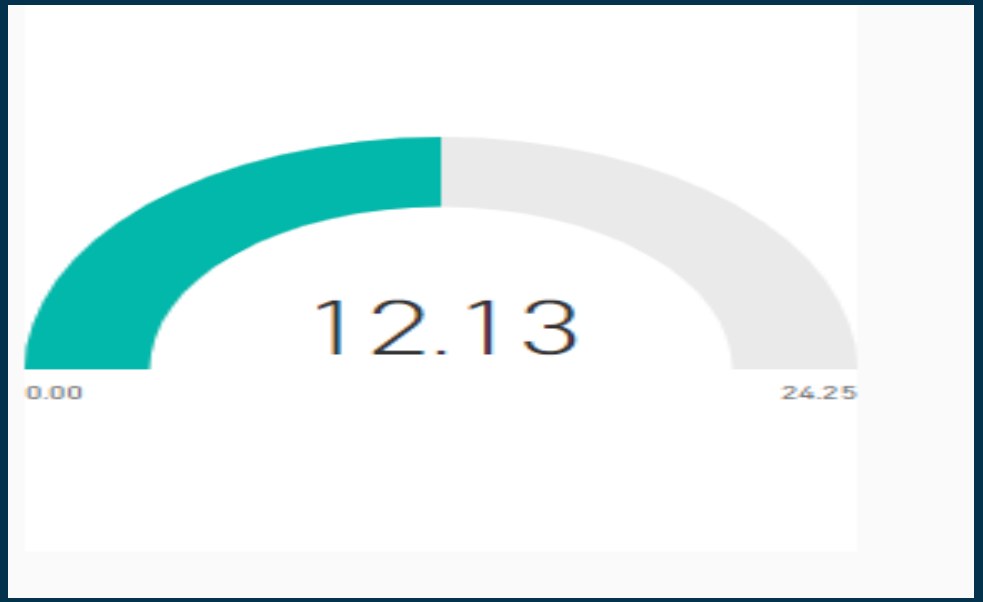
RESULTS-

Real-time Attention and Brain Waves



The real-time monitoring label in Figure shows the absolute values of attention and brain wave amplitudes at the current time section.

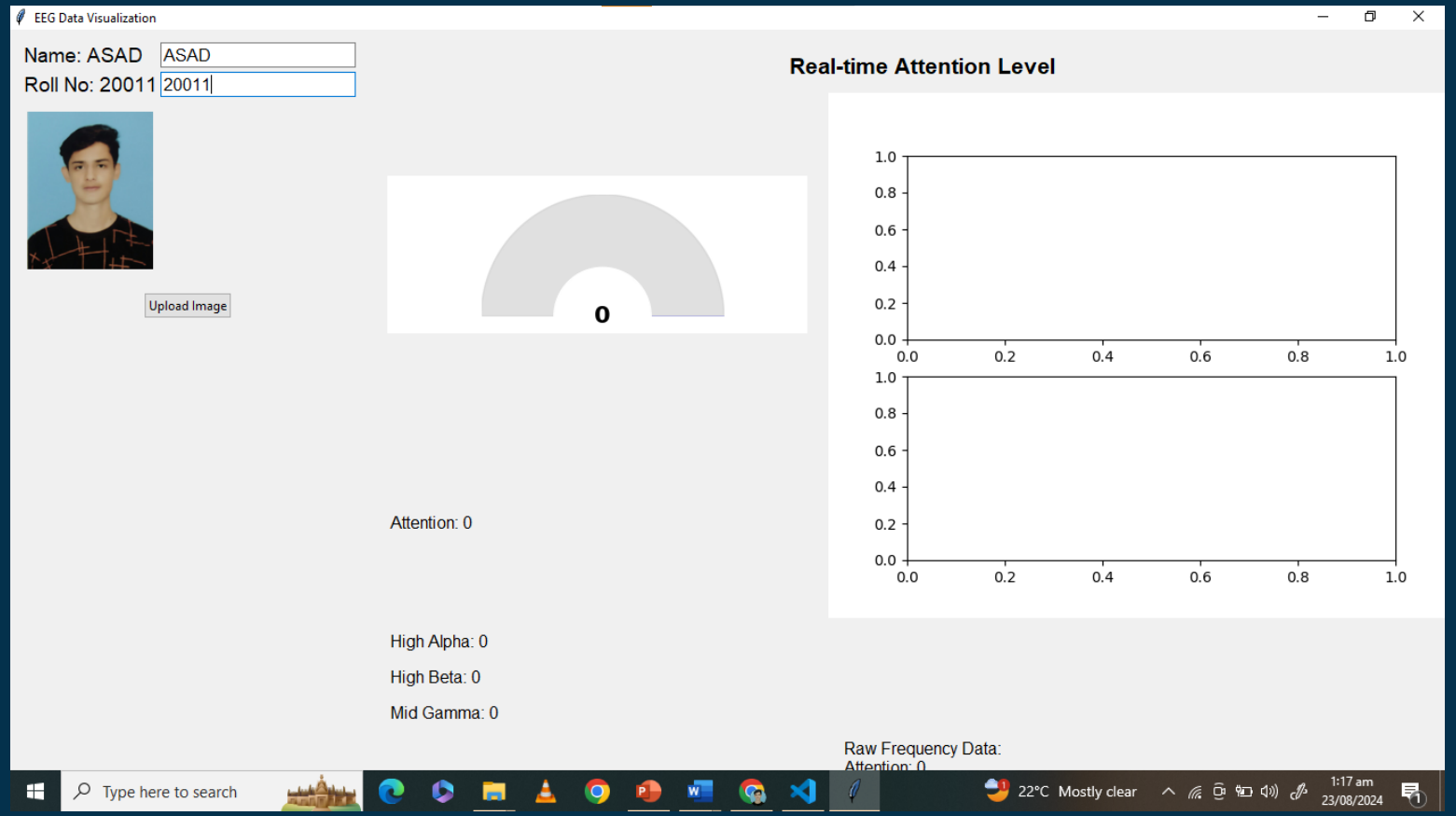
REALTIME EEG FREQUENCY USING PYTHON



This graph is the gauge chart which is designed to illustrate the real time attention level. The gauge ranges from 0.00 to 24.25, with the current attention level marked at 12.13

RESULTS-

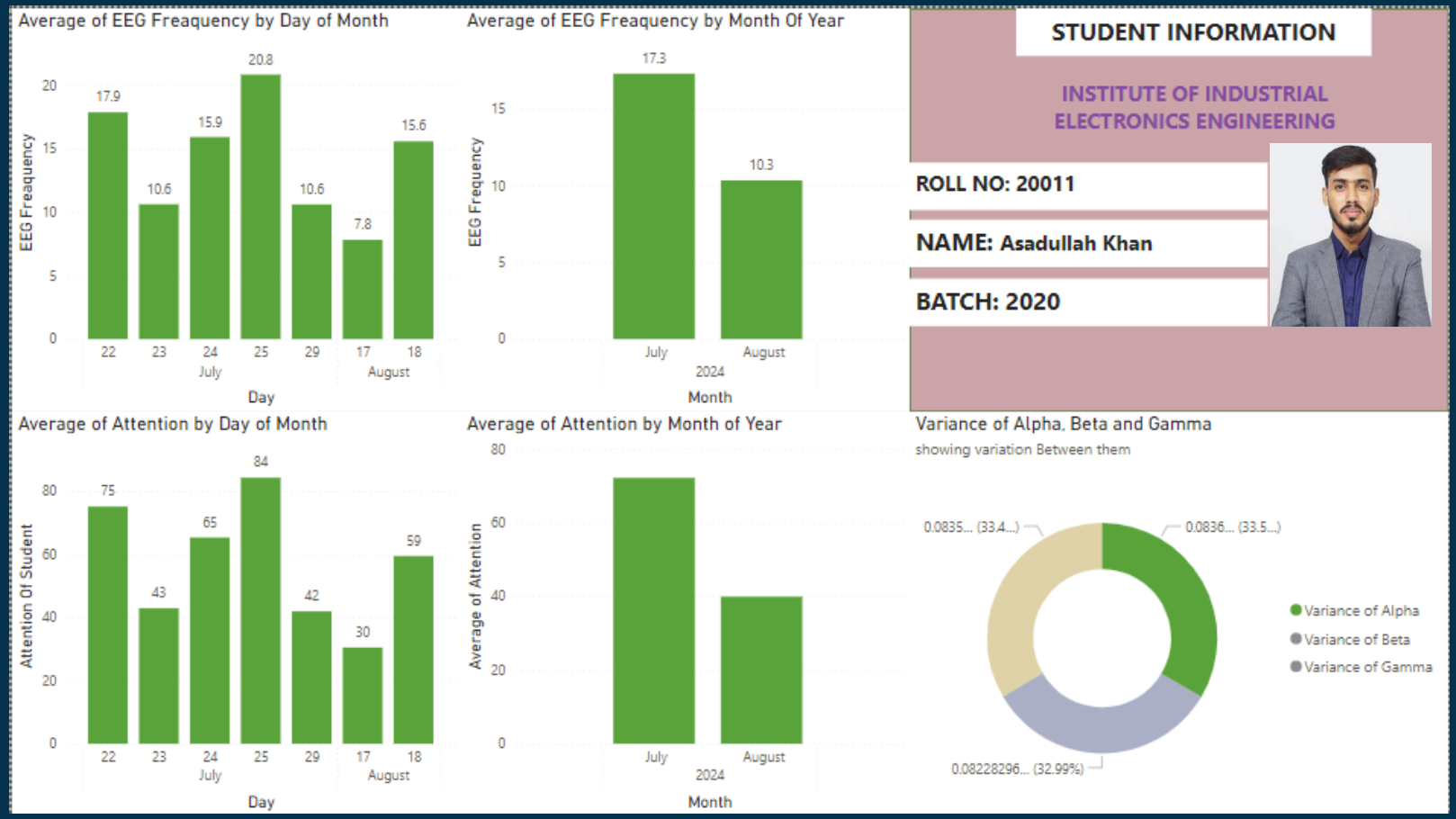
GUI INTERFACE



This shows the proper guy Of Personal Information, Consist of individual performance, graphs, and also attention levels

RESULTS-

Powerbase Dashboard



RESULTS-

PowerBI Dashboard

Showing Student Information.

STUDENT INFORMATION

INSTITUTE OF INDUSTRIAL ELECTRONICS ENGINEERING



ROLL NO: 20011

NAME: Asadullah Khan

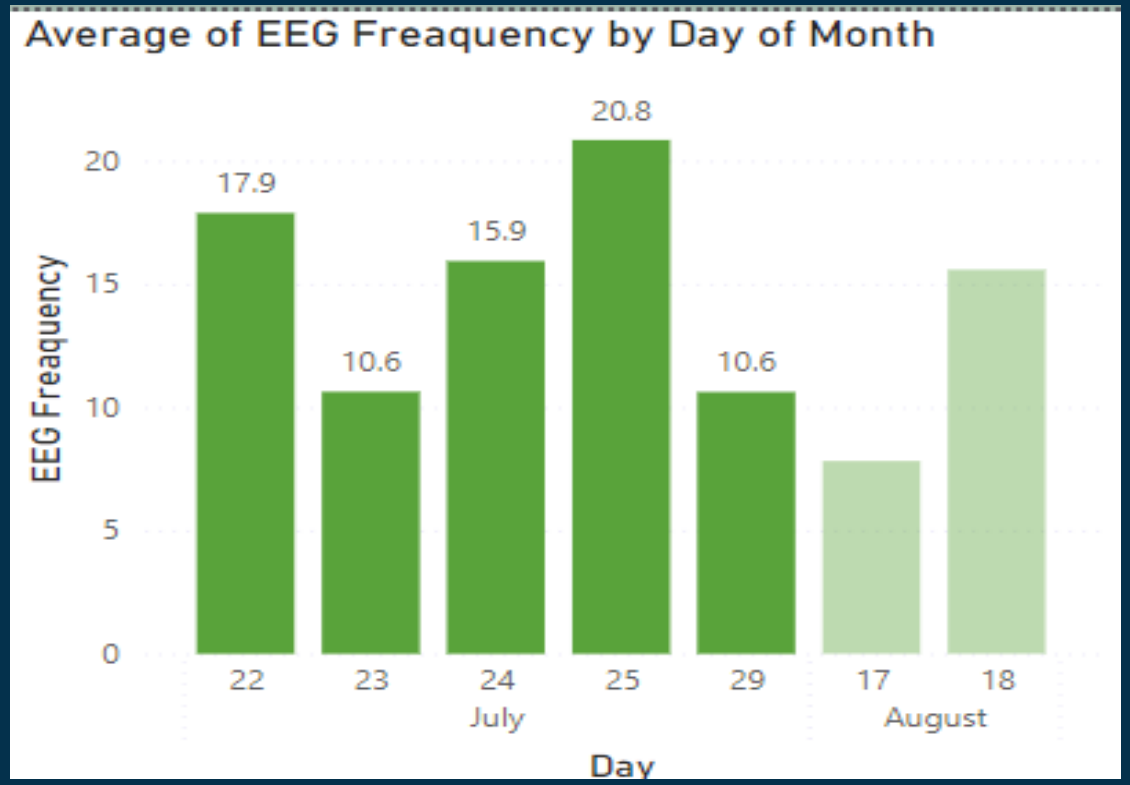
BATCH: 2020

This contains all the info regarding to the student entry, there name roll number and also attached pic

RESULTS-

PowerBI Dashboard

Here column chart showing daily average of EEG Frequency.

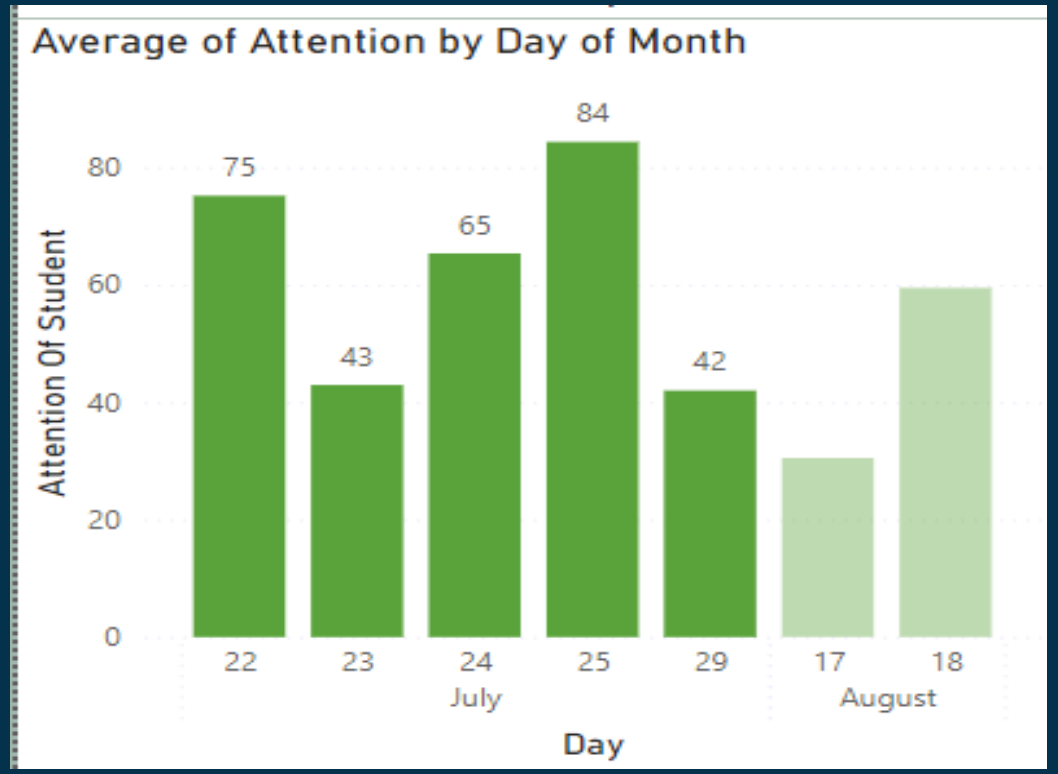


This graph shows the EEG Frequency 17.9 Hz on 22 July shows the mind is in Beta state.

RESULTS-

PowerBI Dashboard

Here column chart showing daily average of Attention.

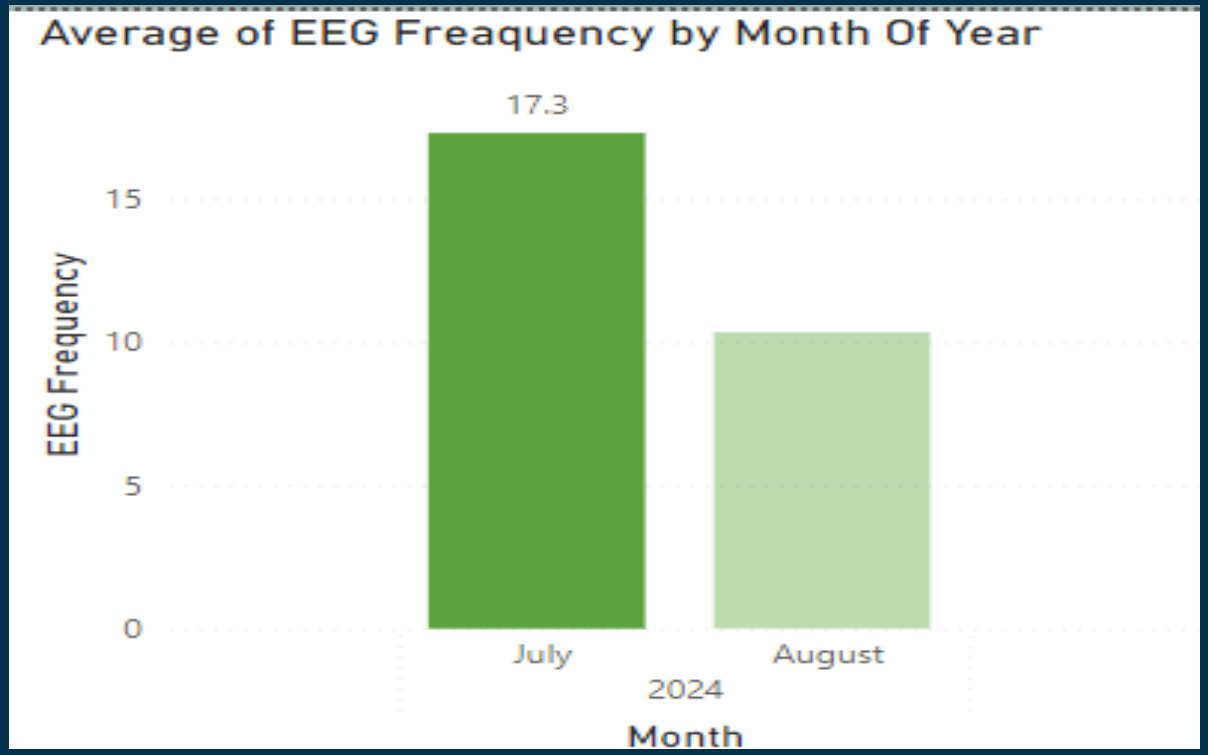


This graph shows the EEG Frequency
75% on 22 July shows the mind is in Beta state.

RESULTS-

PowerBI Dashboard

Here column chart showing Monthly average of EEG Frequency.

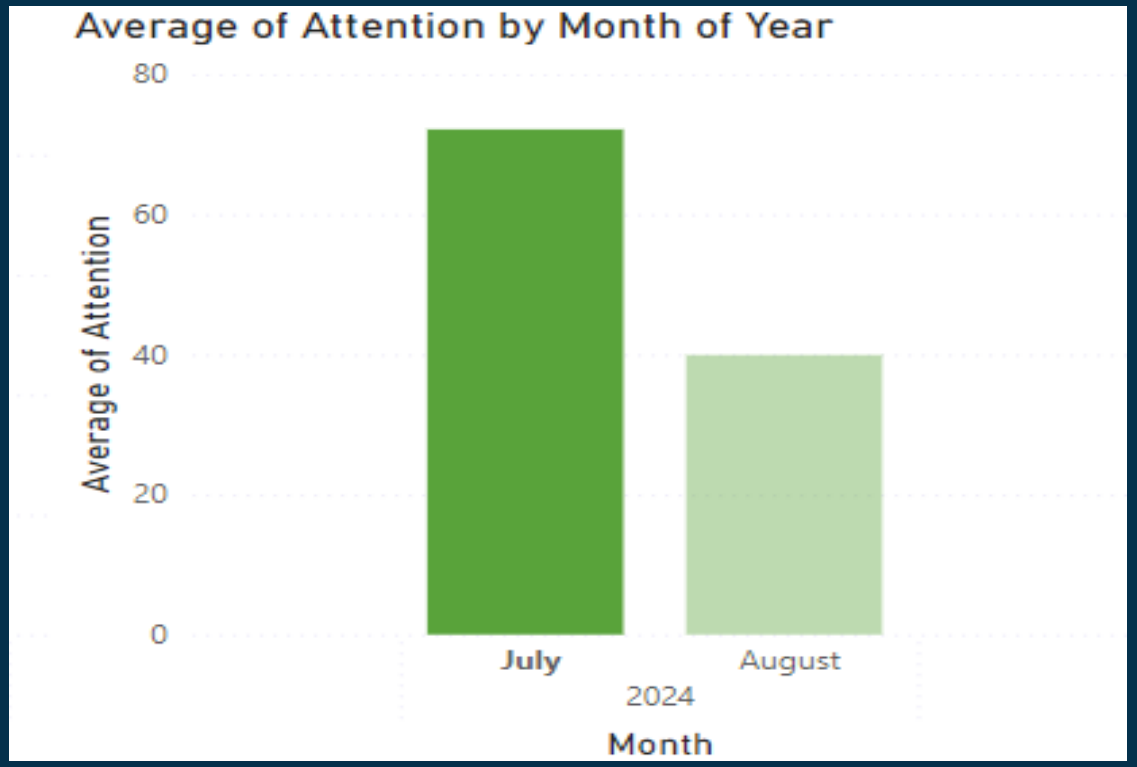


This graph shows the EEG Frequency 17.3 Hz monthly average of July shows the mind state.

RESULTS-

PowerBI Dashboard

Here column chart showing Monthly average of Attention.



This graph shows the EEG Frequency 75% monthly average of July shows the mind state.

RESULTS-

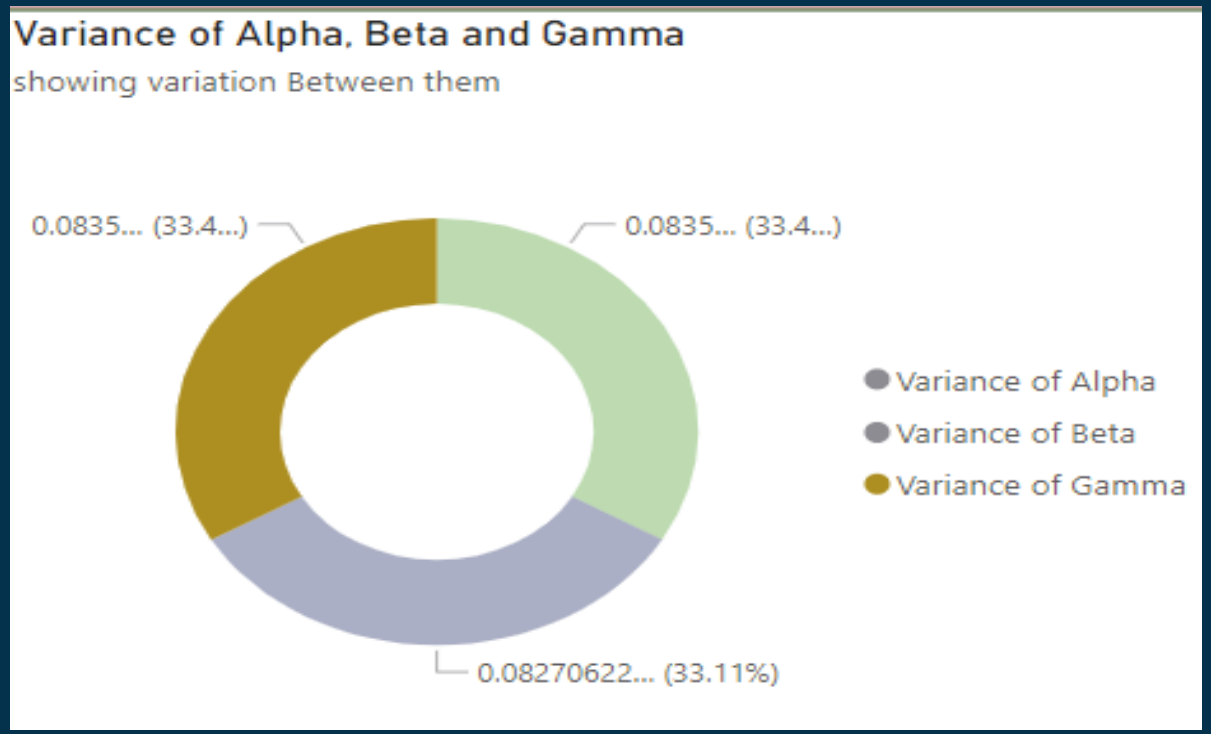
PowerBI Dashboard

Donut chart showing variance of alpha beta gamma, showing variation in them.

Formula

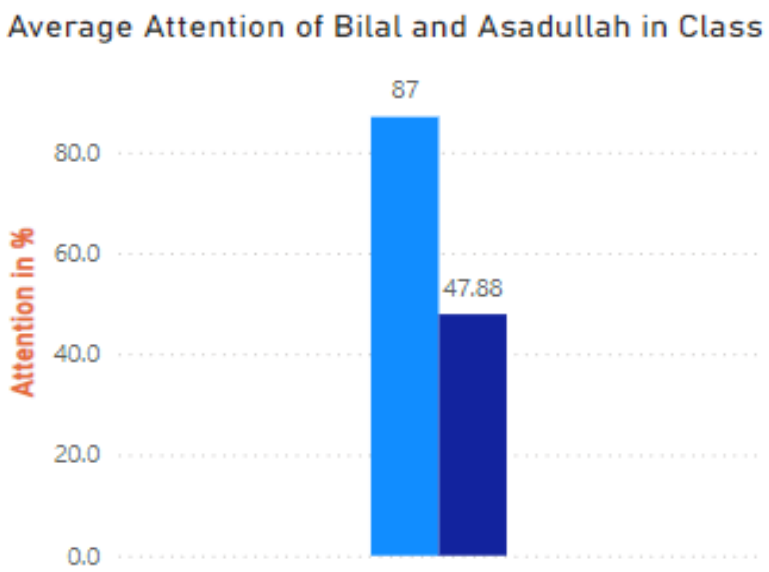
$$S^2 = \frac{\sum(x_i - \bar{x})^2}{n - 1}$$

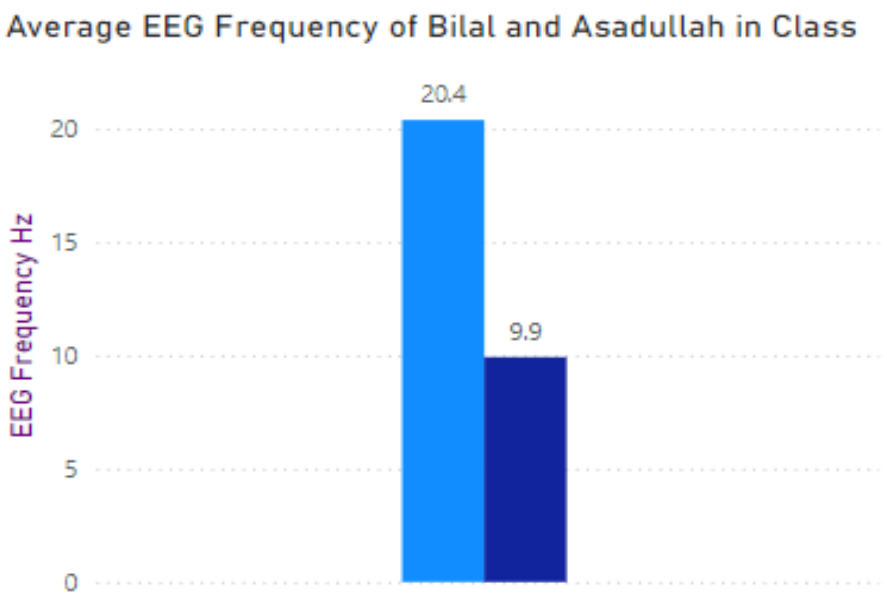
S^2 = sample variance
x_i = the value of the one observation
$\bar{x}$ = the mean value of all observations
n = the number of observations



RESULTS-

Comparison of Two Students in Class

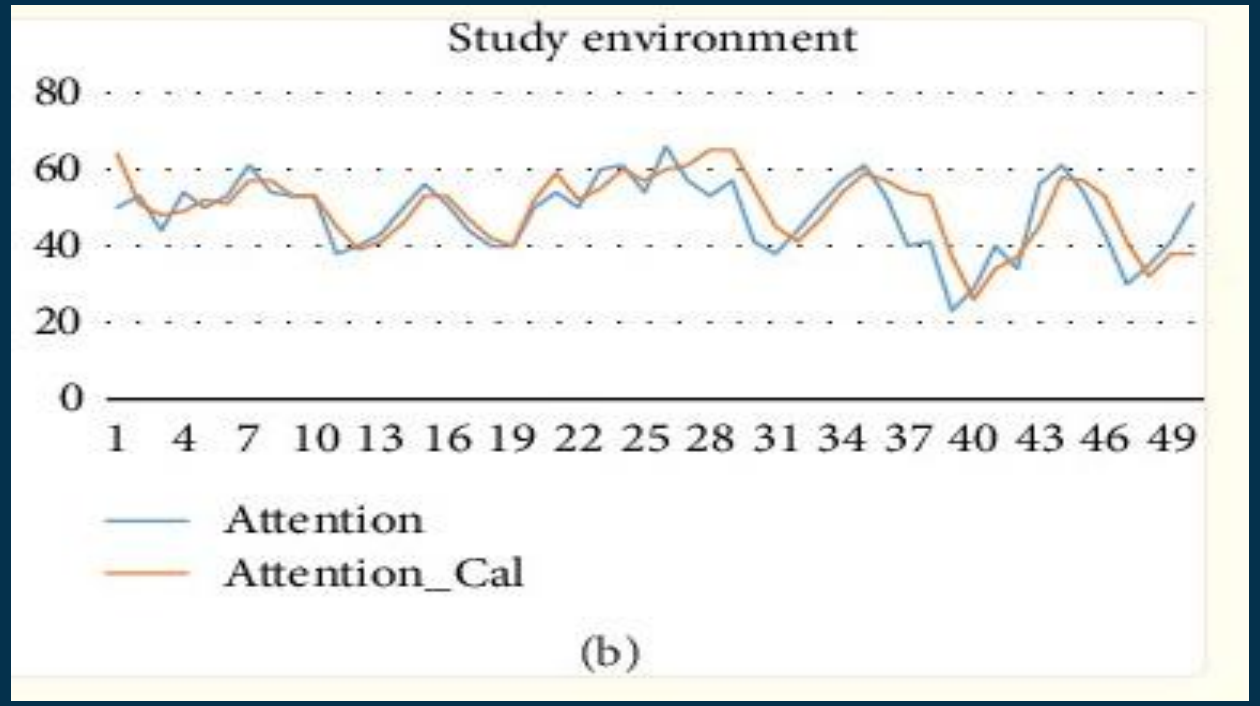




This is the comparison of the two students Bilal and Asadullah taking a class of 1 hour same subject.
Here we can see the difference between both of them

RESULTS-

Comparisons of Attention In Waveform



This is the comparison of the two students Bilal and Asadullah taking a class of 1 hour same subject.
We are comparing 50 initial samples